



COMPUTER VISION

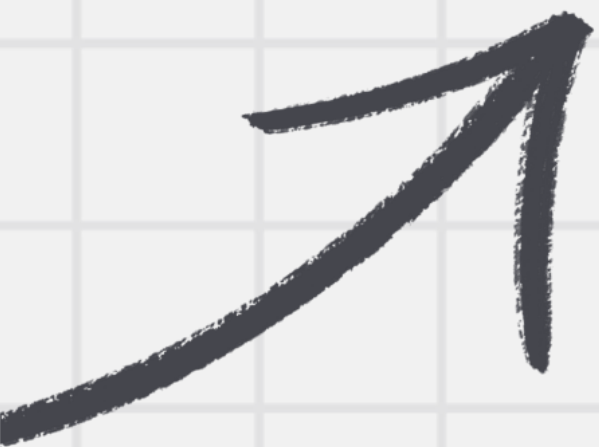
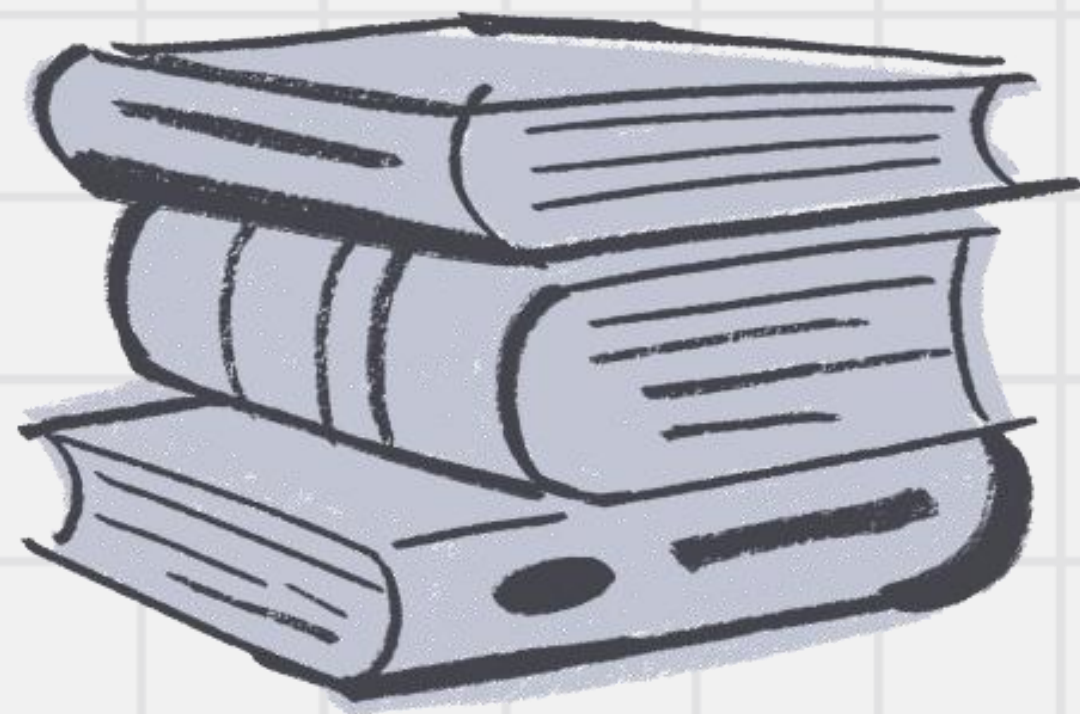


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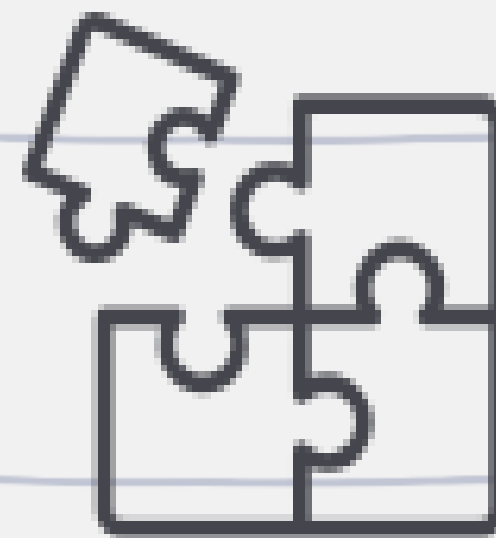
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01



Introduction

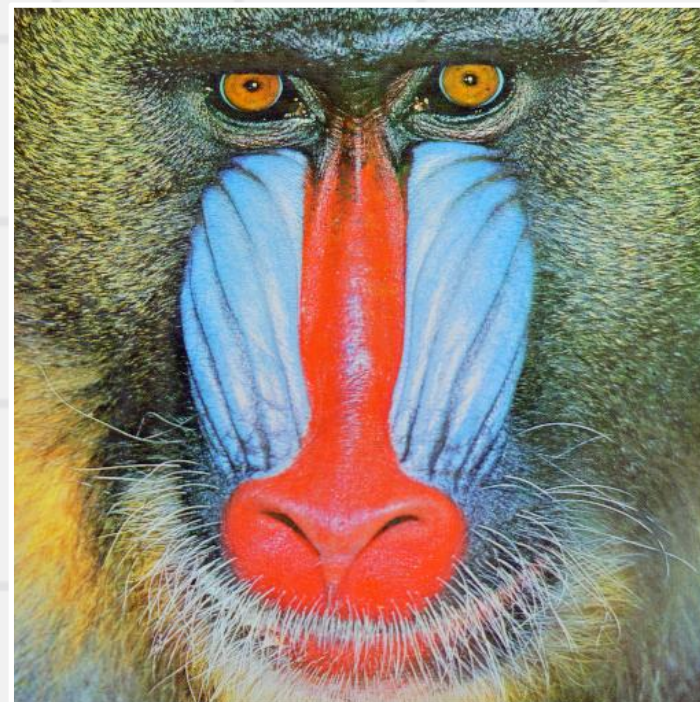
Introduction

Last time, we finished our talk with a question. can you think of alternative methods to dilation and erosion for enhancing object boundaries and removing noise in binary images? 🤔💡

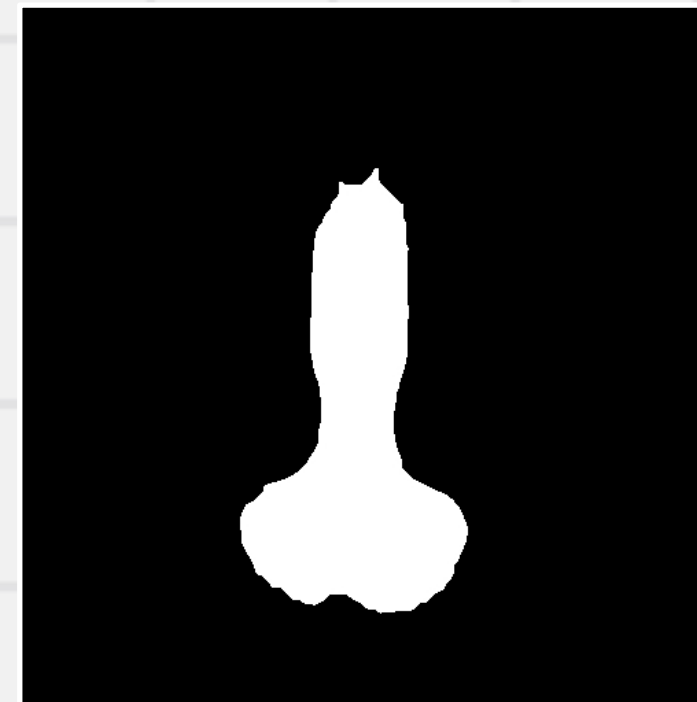
The answer is yes, simply that would happen by two stages:

1. Apply FFT instead of Convolution Blur
2. Apply closing or opening instead of Dilation and Erosion

Original Image



Final Image





02



Fourier Series

Fourier Series

The Fourier transform basically says that any periodic function can be written as a weighted sum of infinite sinusoids of different frequencies.

The sinusoid formula is like the following:

$$f(x) = A \sin(2\pi u x + \varphi)$$

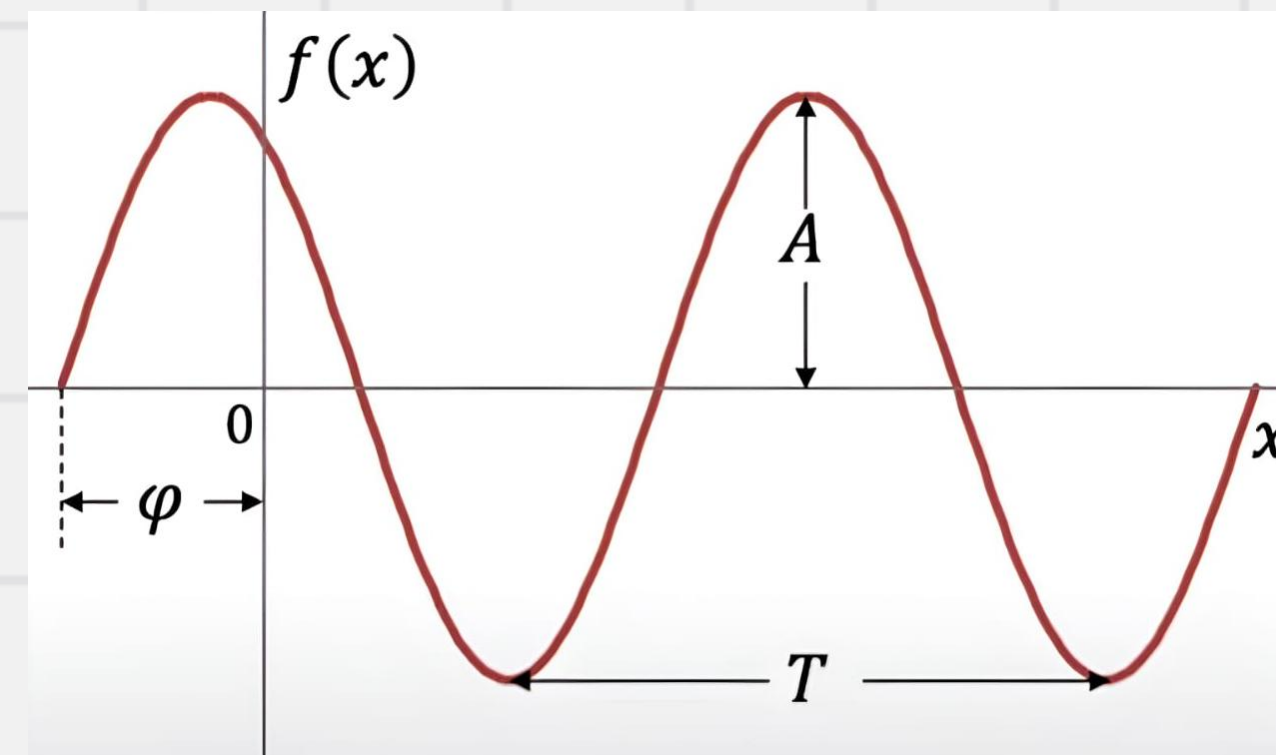
Which

A : Amplitude

T : Period

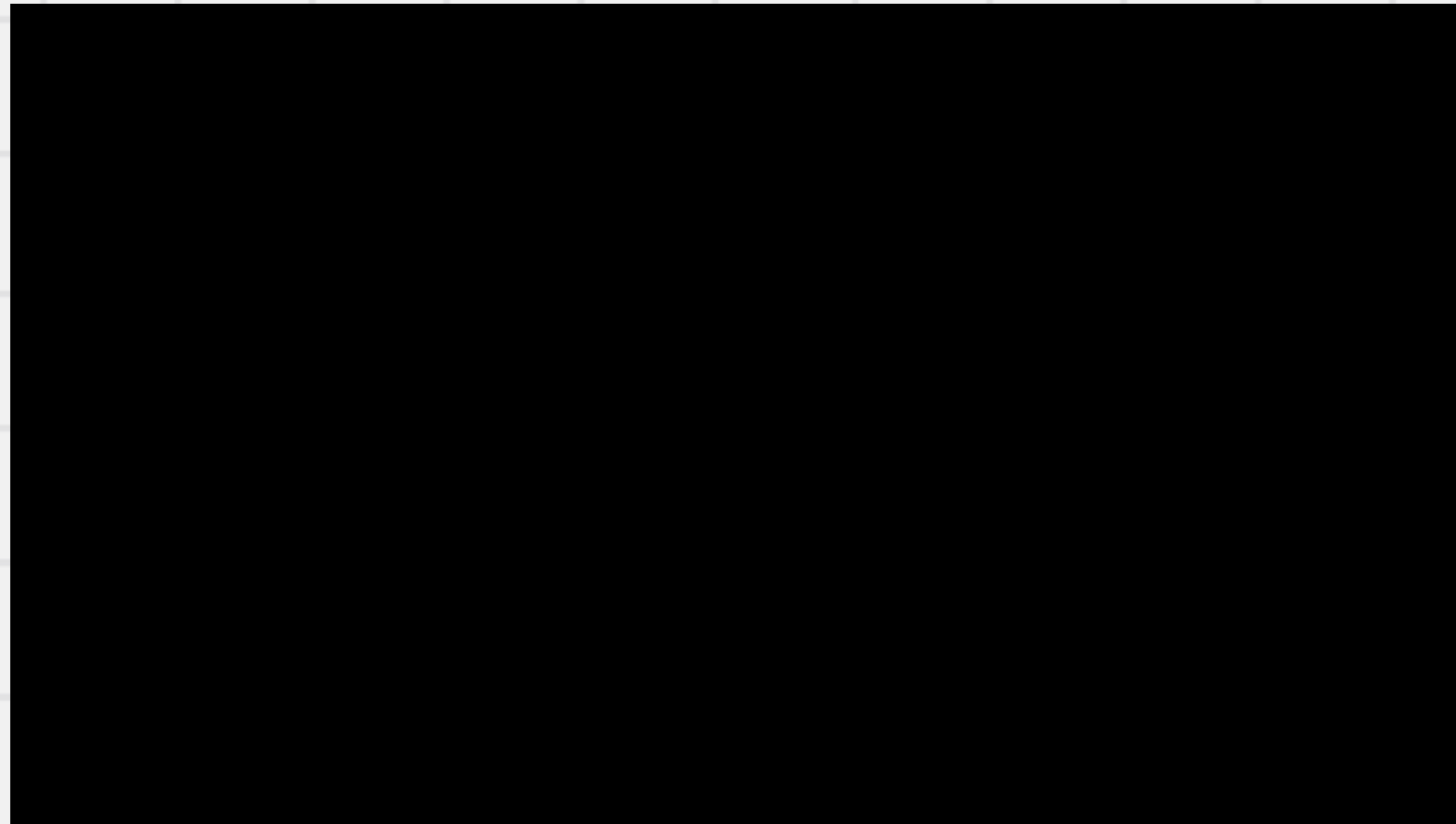
φ : Phase

u : Frequency ($\frac{1}{T}$)



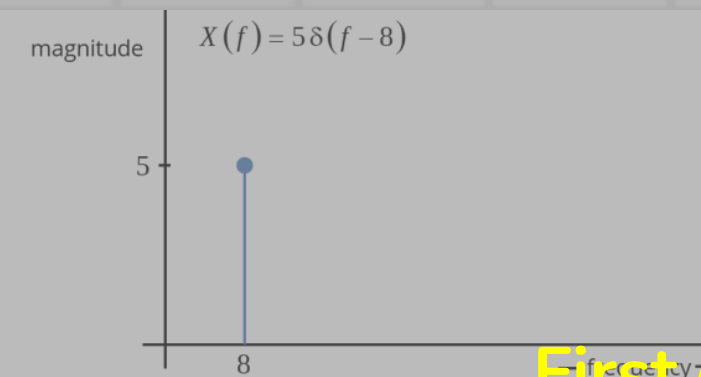
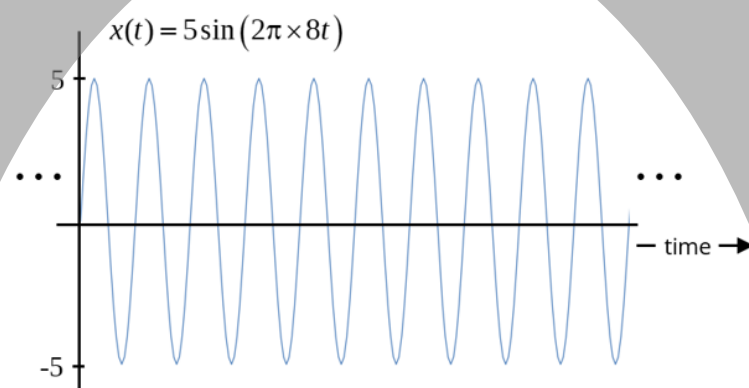
Fourier Series

As we said that any periodic function can be described using infinite sum of sinusoids. Look how can we generate square wave using Fourier series.

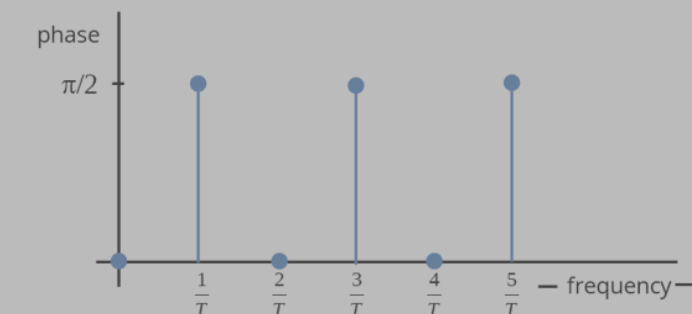
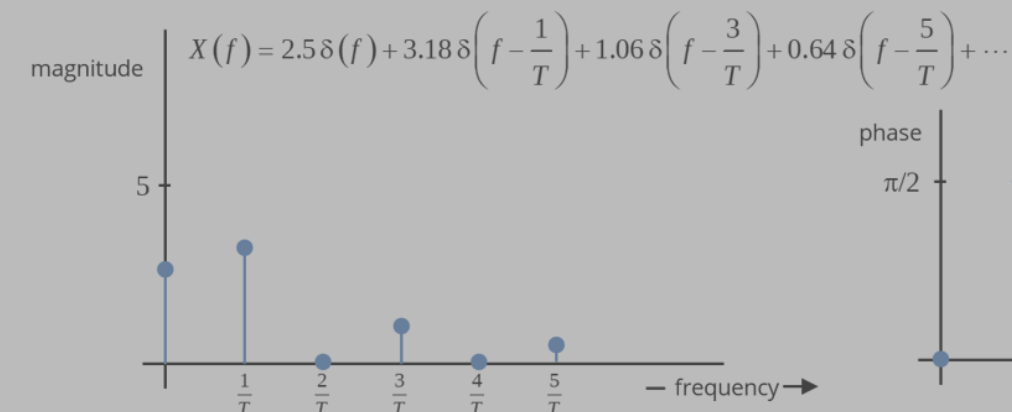
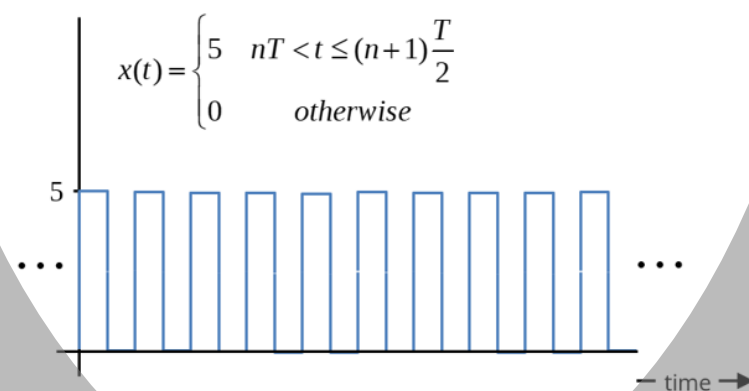
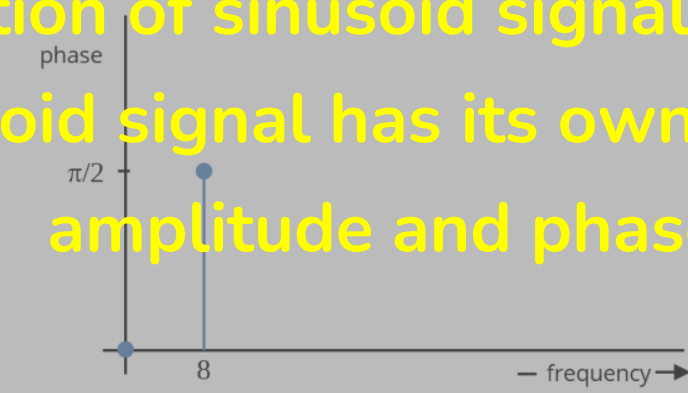
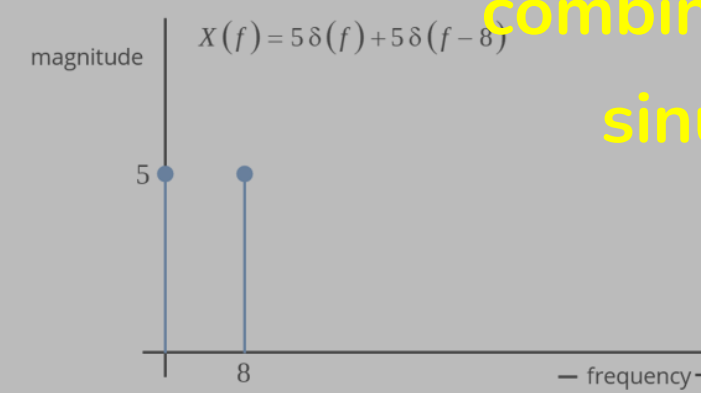
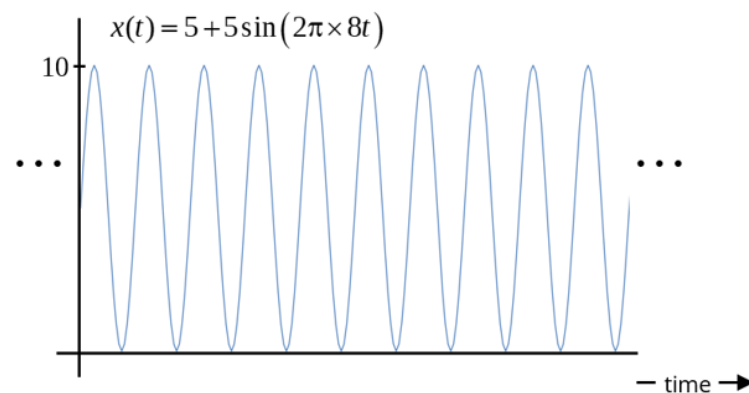


Fourier Series

Now, what can we infer from that? And how can it help us to reduce image noise?



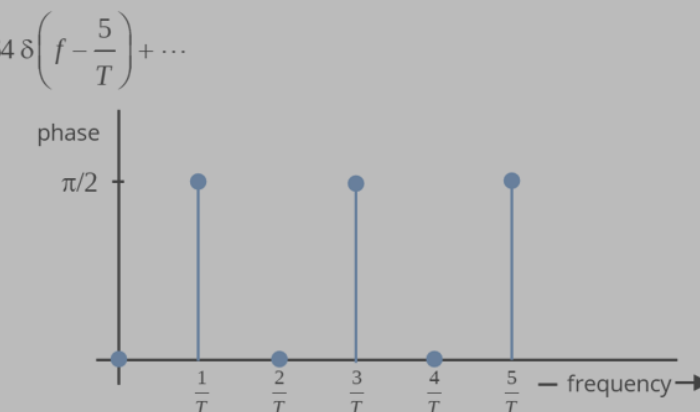
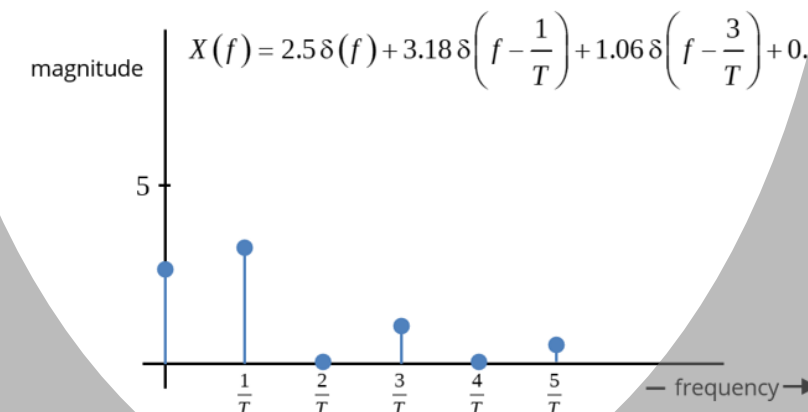
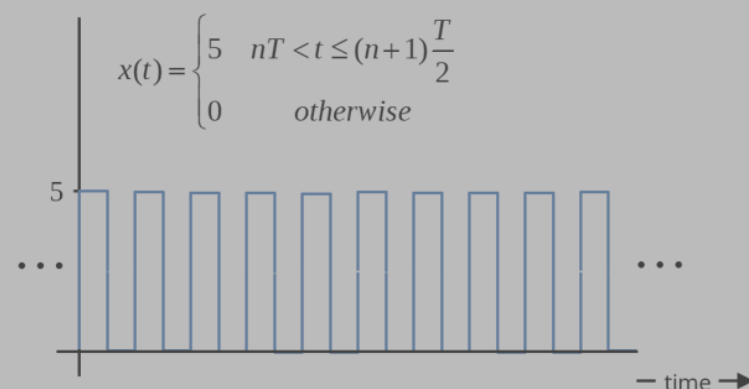
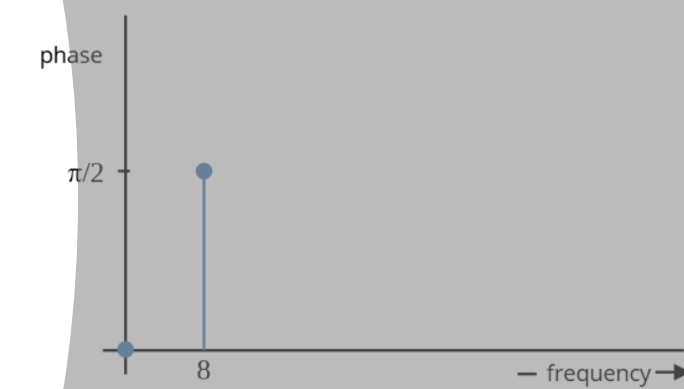
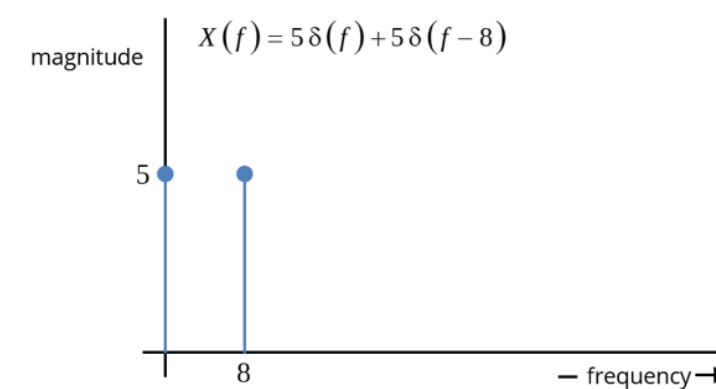
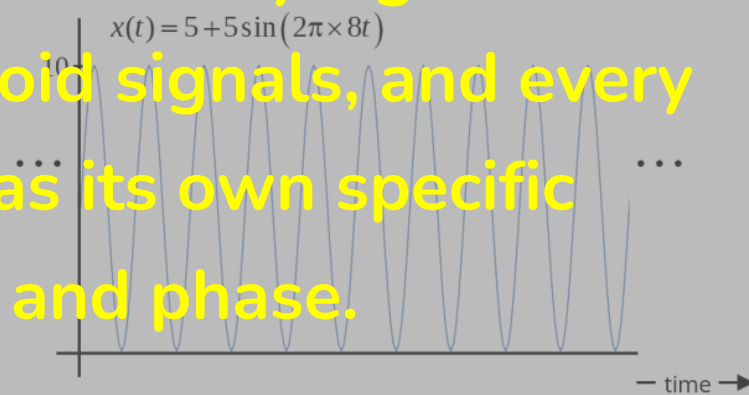
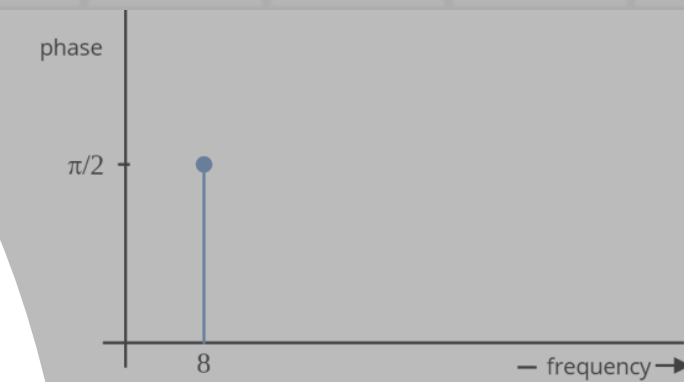
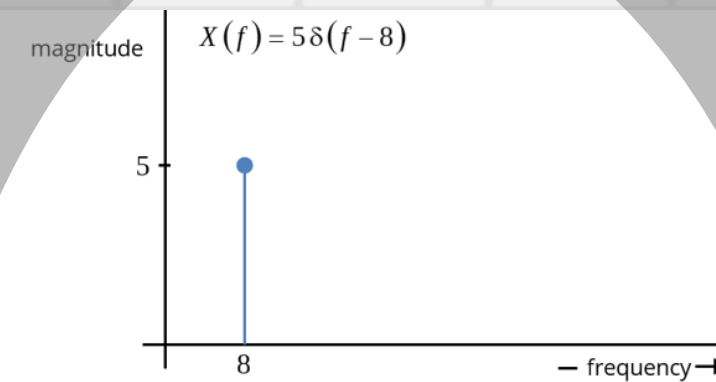
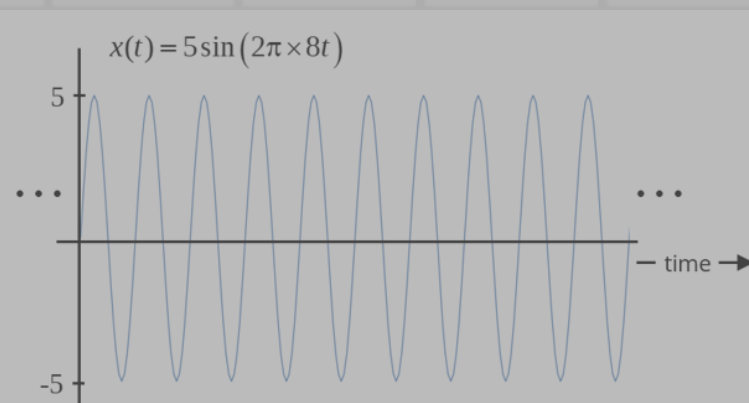
First conclusion is that every signal is a combination of sinusoid signals, and every sinusoid signal has its own specific amplitude and phase.



Fourier Series

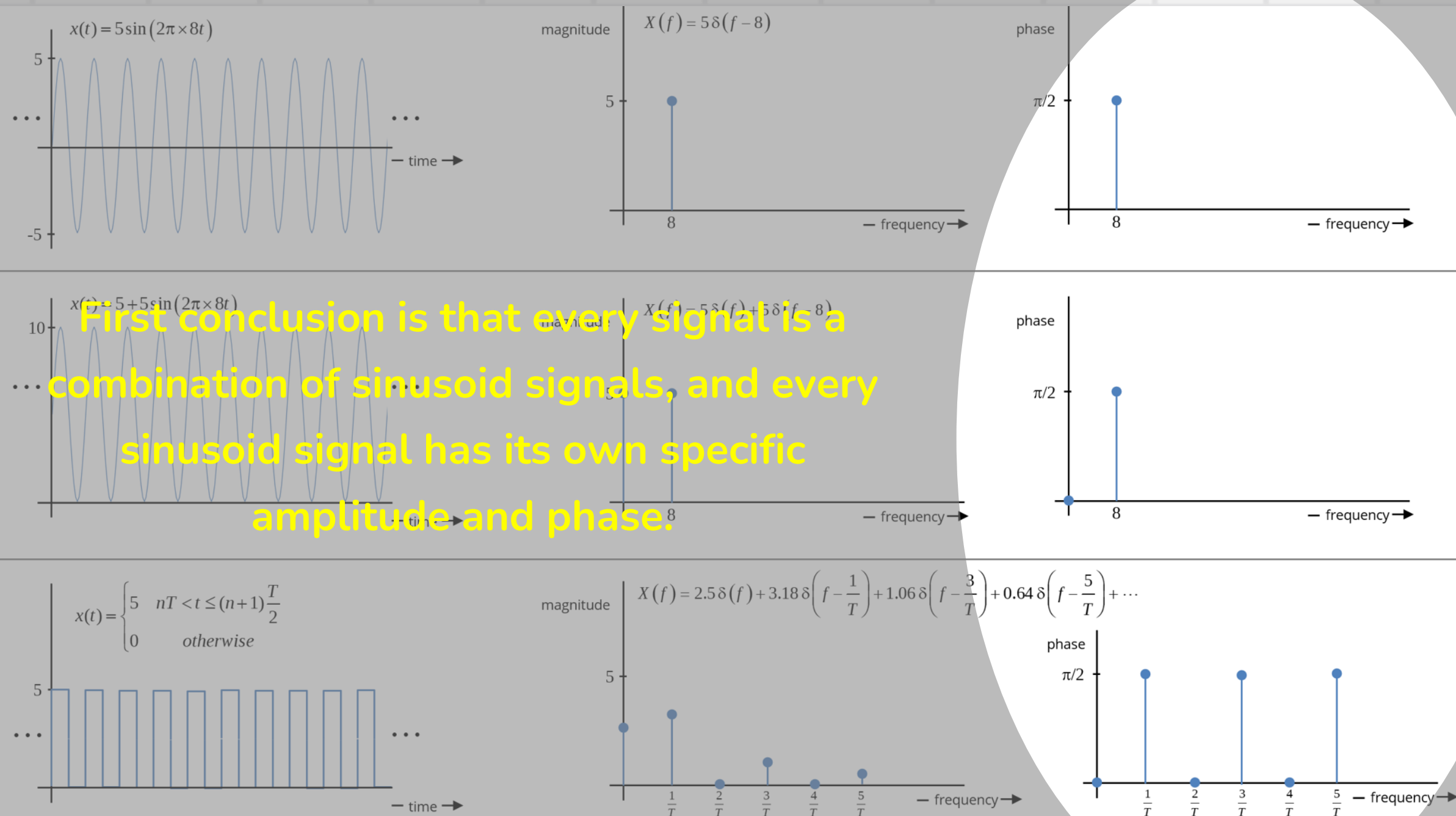
Now, what can we infer from that? And how can it help us to reduce image noise?

First conclusion is that every signal is a combination of sinusoid signals, and every sinusoid signal has its own specific amplitude and phase.



Fourier Series

Now, what can we infer from that? And how can it help us to reduce image noise?

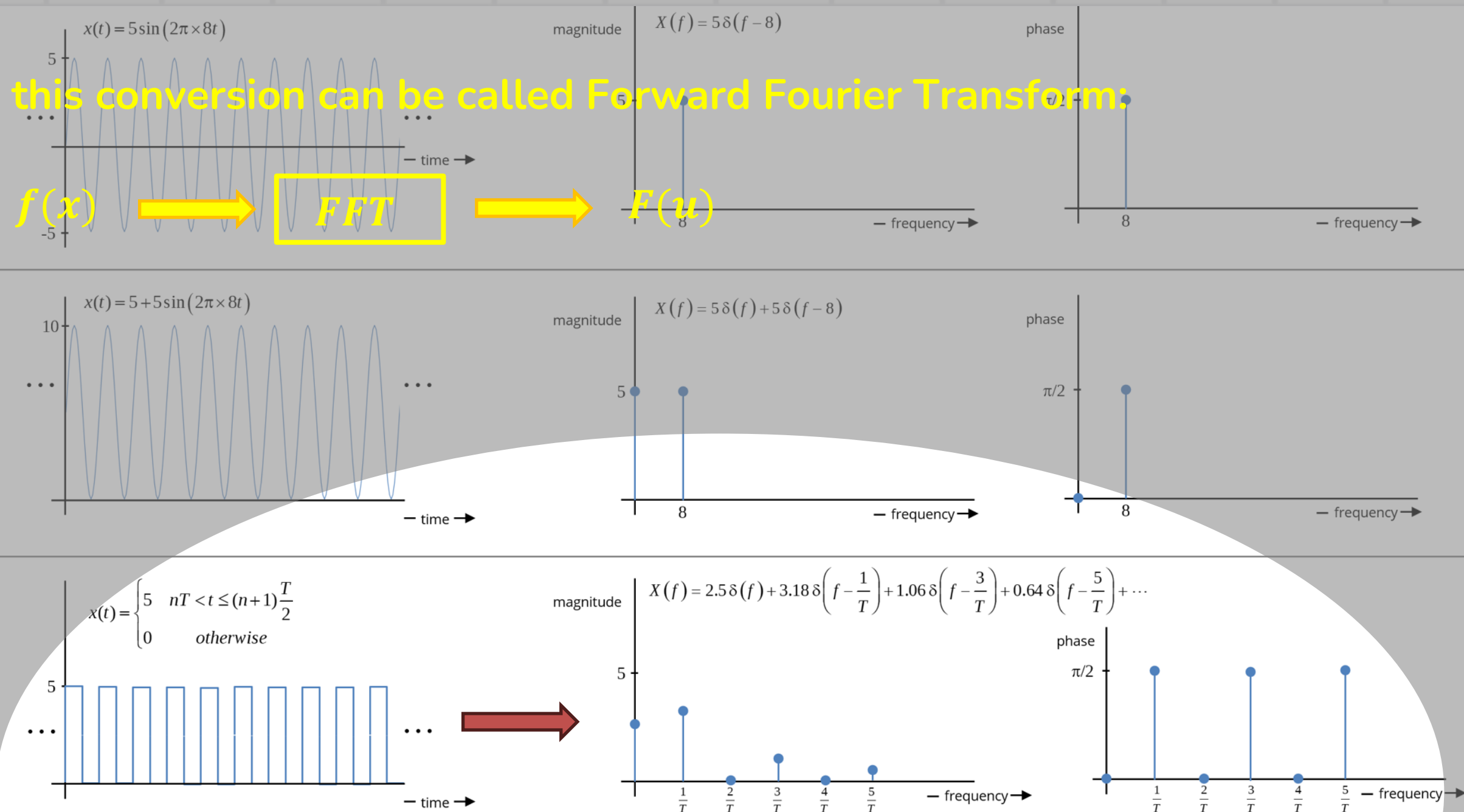


First conclusion is that every signal is a combination of sinusoid signals, and every sinusoid signal has its own specific amplitude and phase.

Fourier Series

Now, what can we infer from that? And how can it help us to reduce image noise?

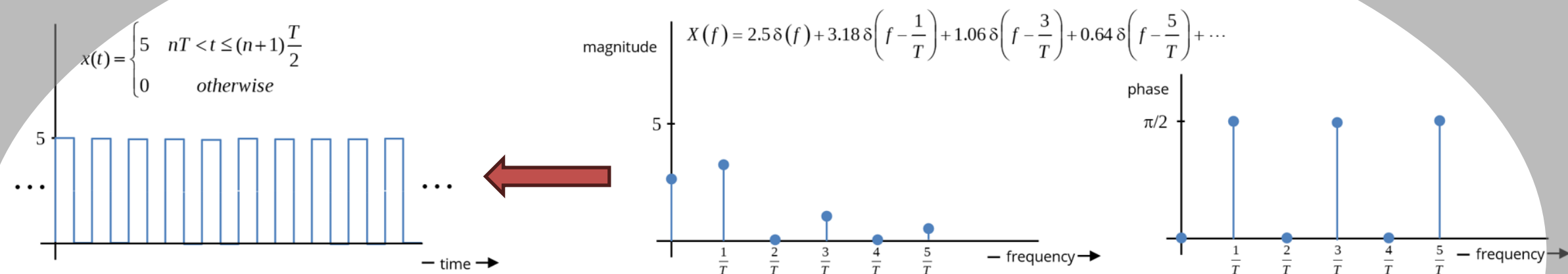
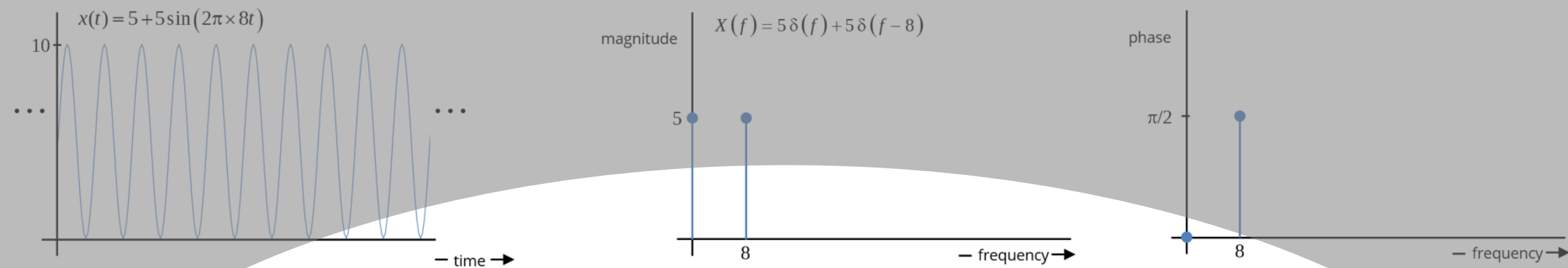
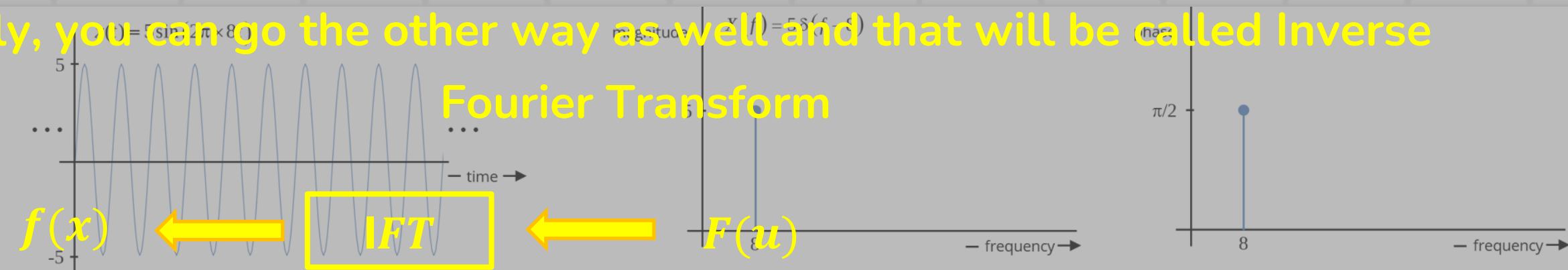
And this conversion can be called Forward Fourier Transform:



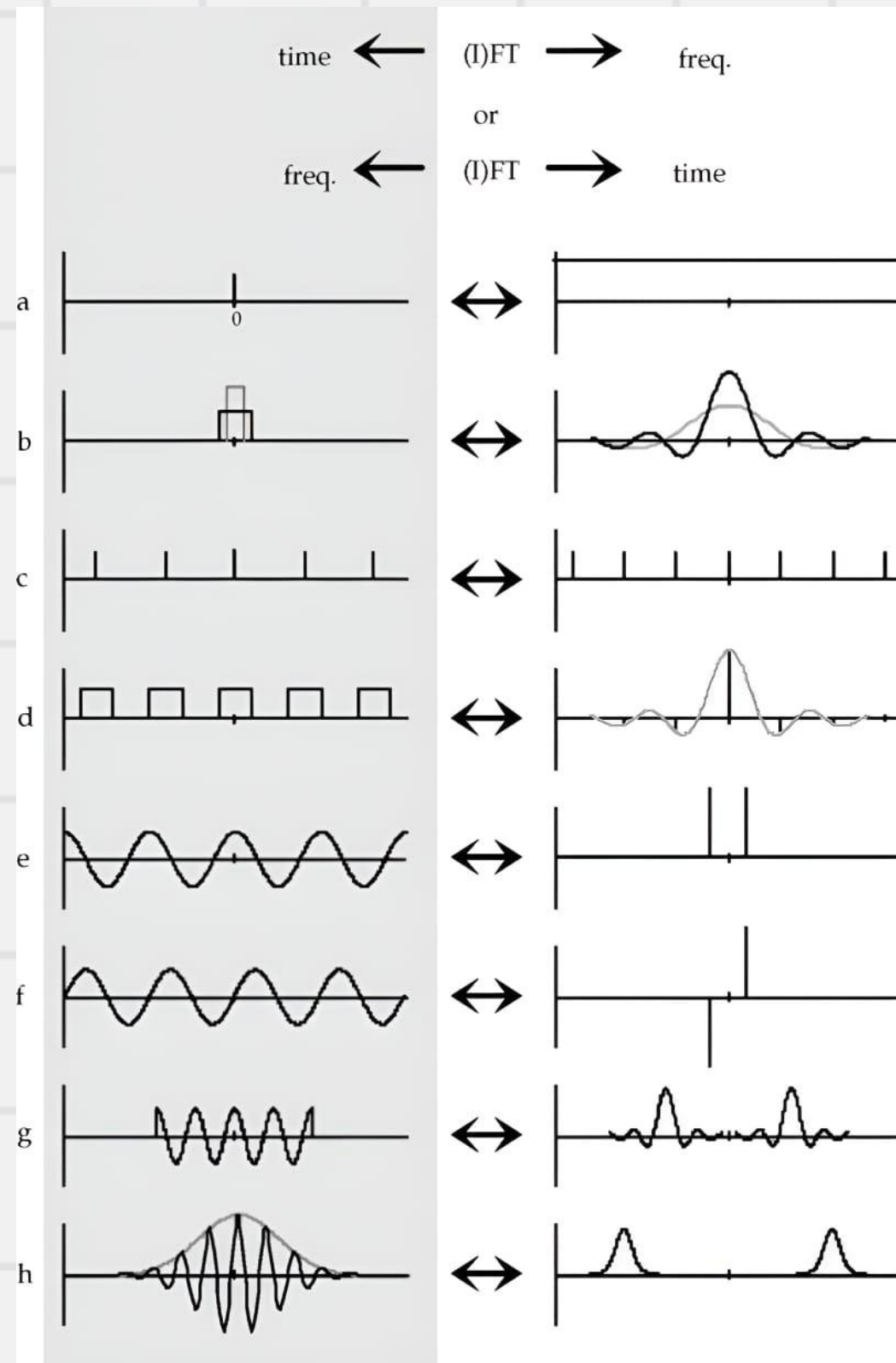
Fourier Series

Now, what can we infer from that? And how can it help us to reduce image noise?

Similarly, you can go the other way as well and that will be called Inverse



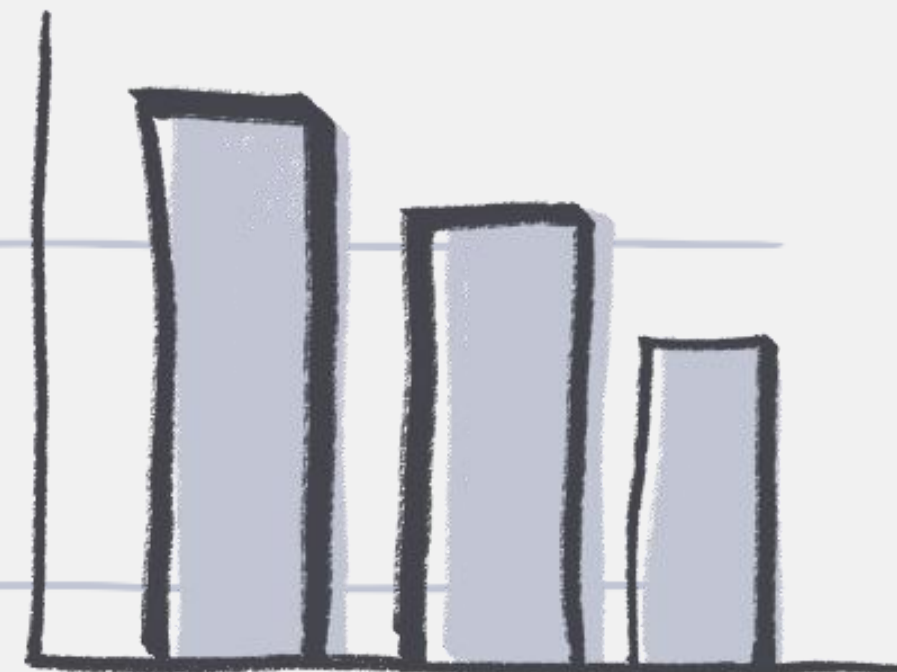
Fourier Series



Look at the different signals and their
Fourier transform



03

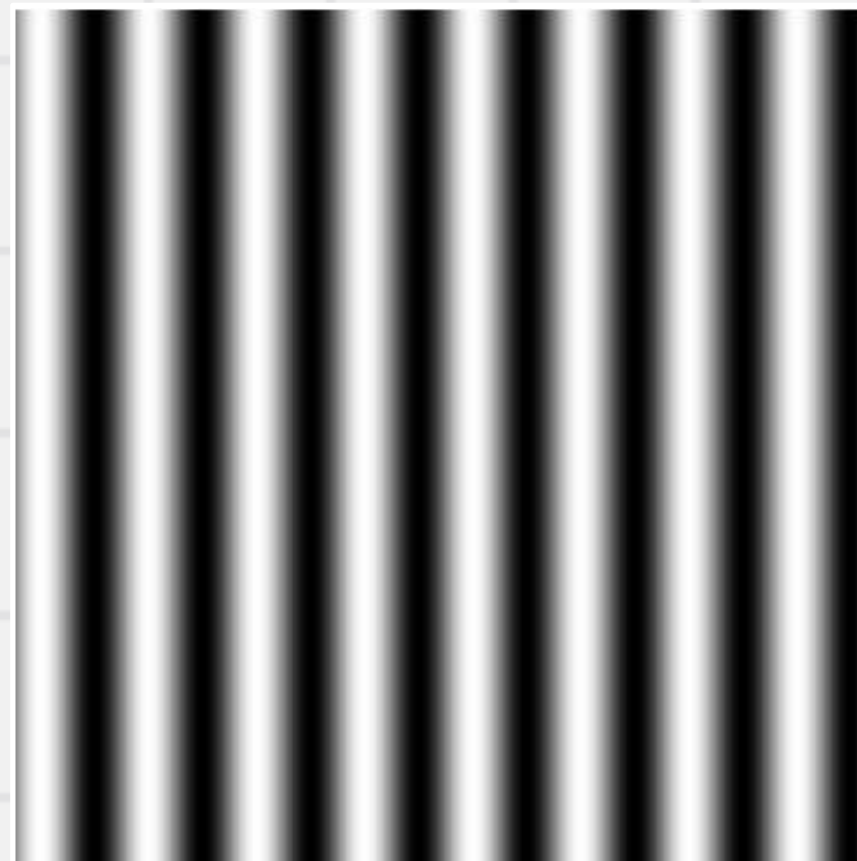


Fast Fourier Transform

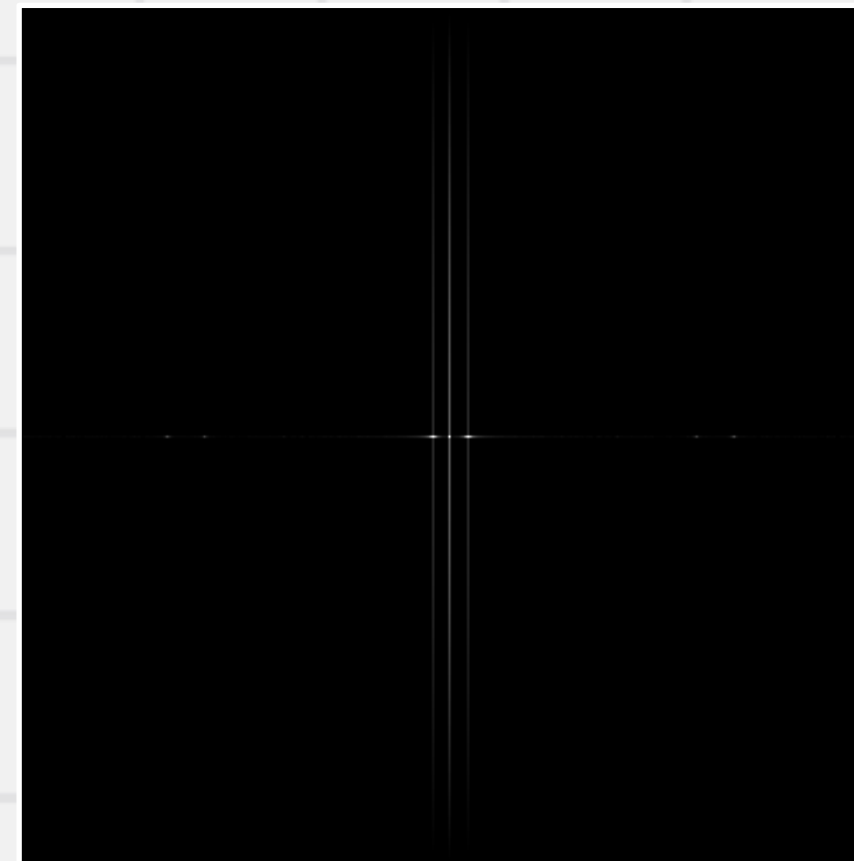
Fast Fourier Transform

Look how this happens in images:

Sine Wave Image



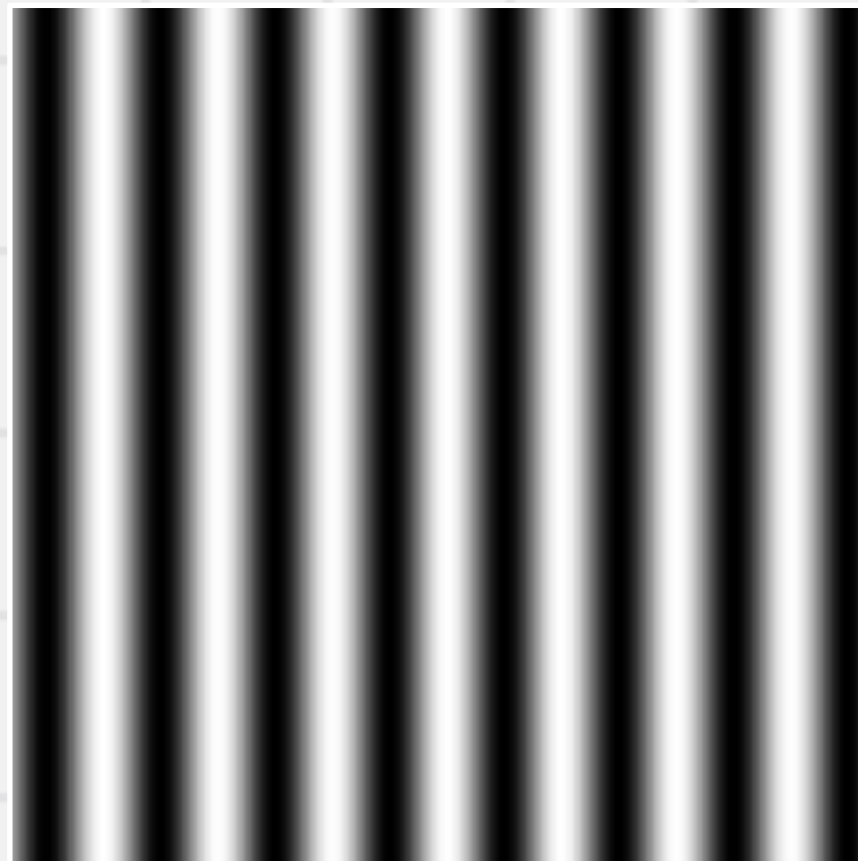
Forward Fourier Sine



Fast Fourier Transform

Look how this happens in images:

Cosine Wave Image



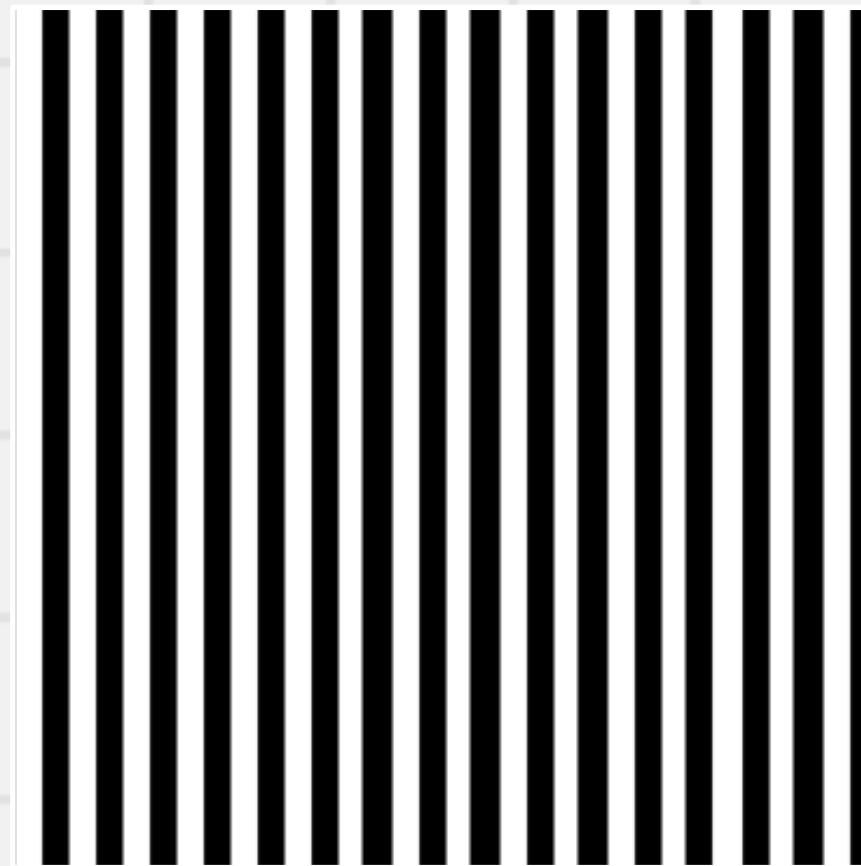
Forward Fourier Cosine



Fast Fourier Transform

Look how this happens in images:

Square Wave Image



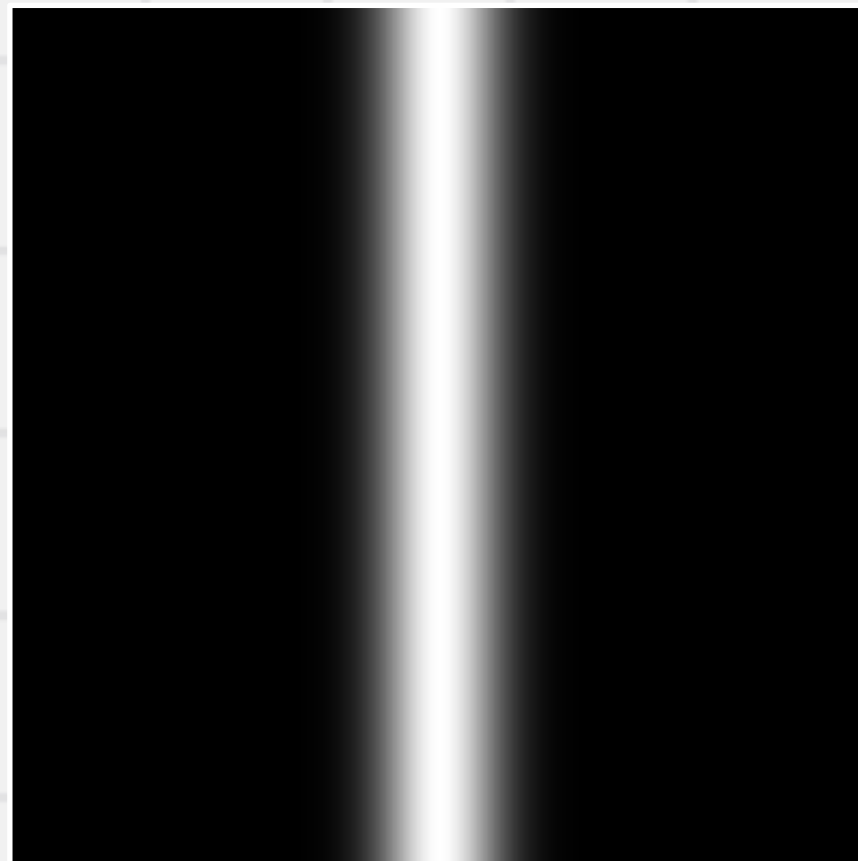
Forward Fourier Square



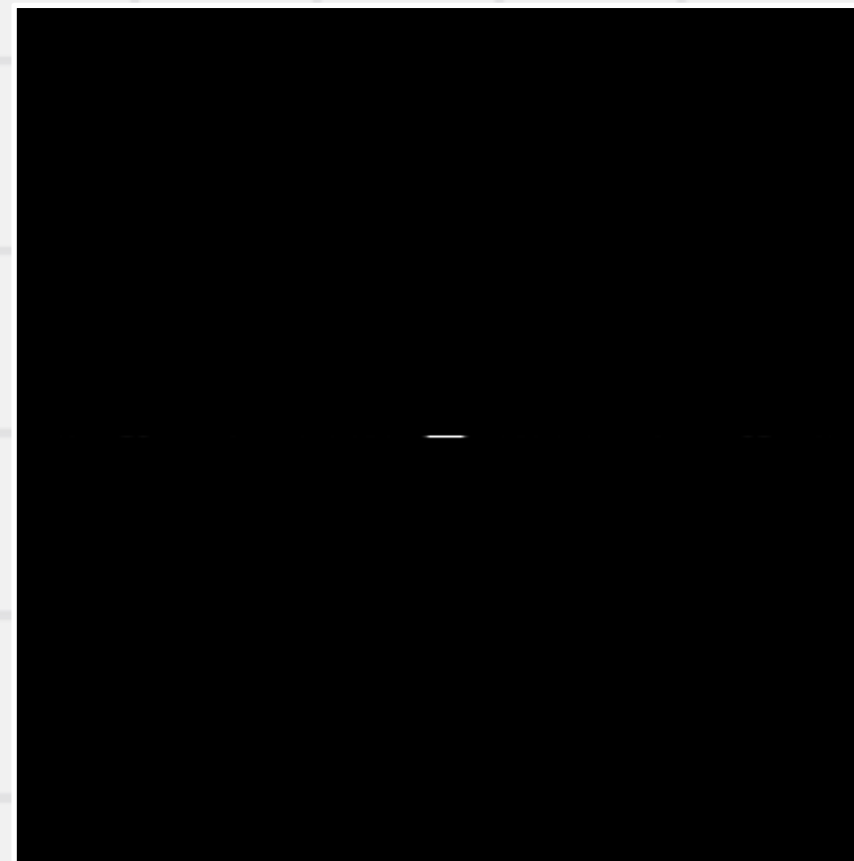
Fast Fourier Transform

Look how this happens in images:

Gaussian Wave Image



Forward Fourier Gaussian

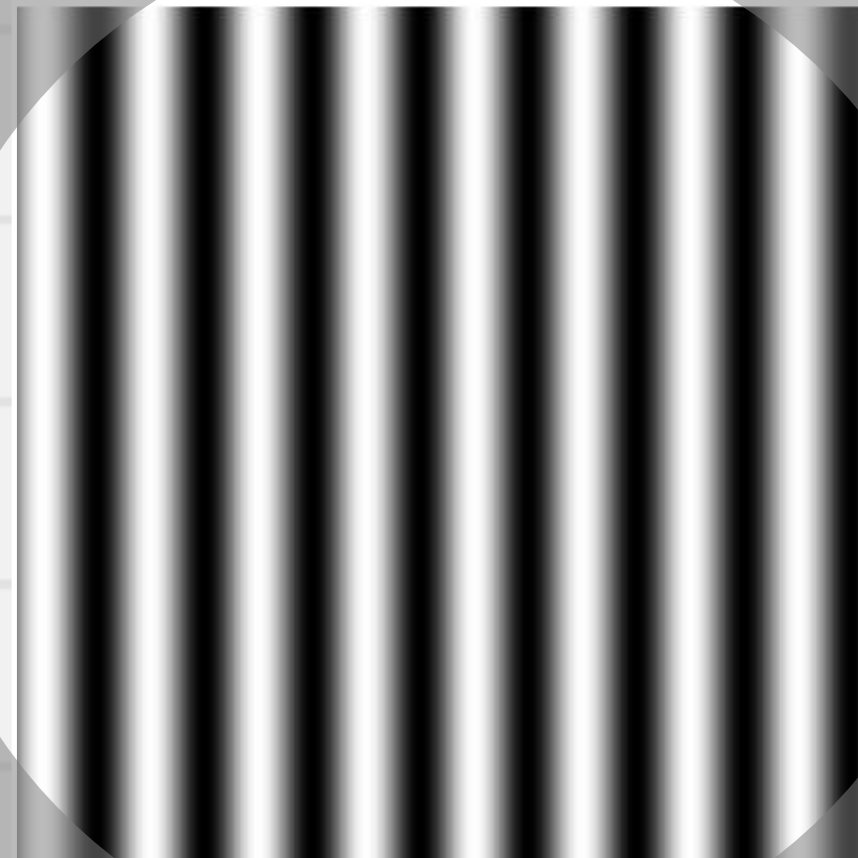


Fast Fourier Transform

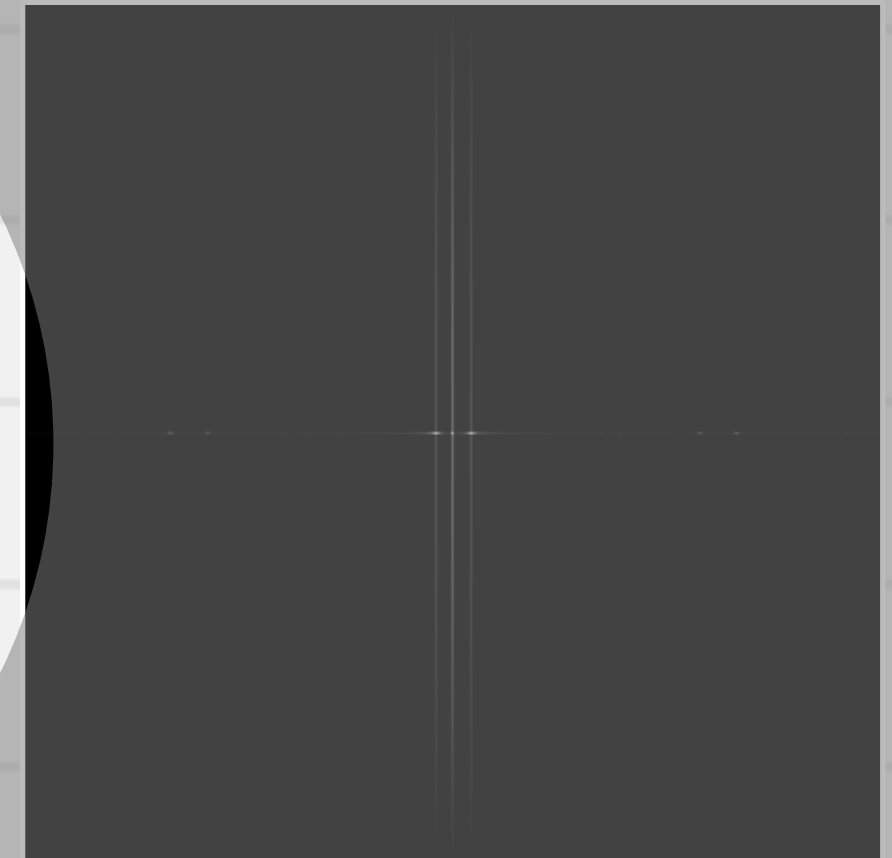
Now let's think how can we use that?

If we want to remove the high frequencies from this image how it can be done?

Sine Wave Image



Forward Fourier Sine



Fast Fourier Transform

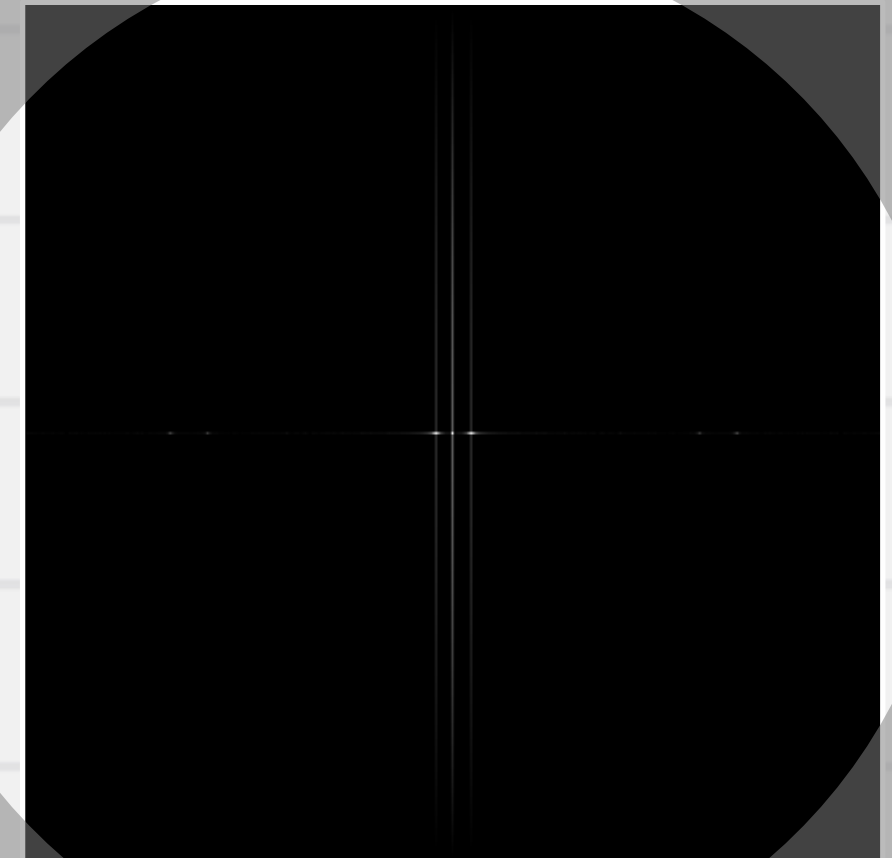
Now let's think how can we use that?

Simply, we will apply low pass filter which leaves low frequencies pass and the higher ones blocked.

Sine Wave Image



Forward Fourier Sine



Fast Fourier Transform

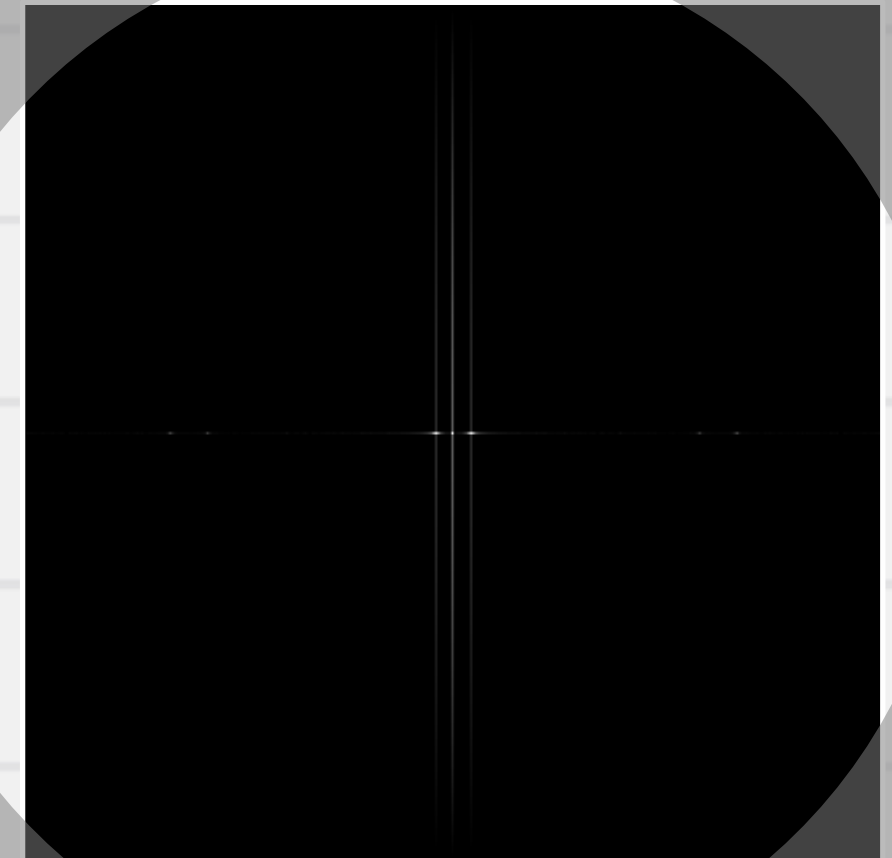
Now let's think how can we use that?

And this will be visualized in forward fourier image as cropping the image center and remove all the pixels which are surrounding it.

Sine Wave Image



Forward Fourier Sine



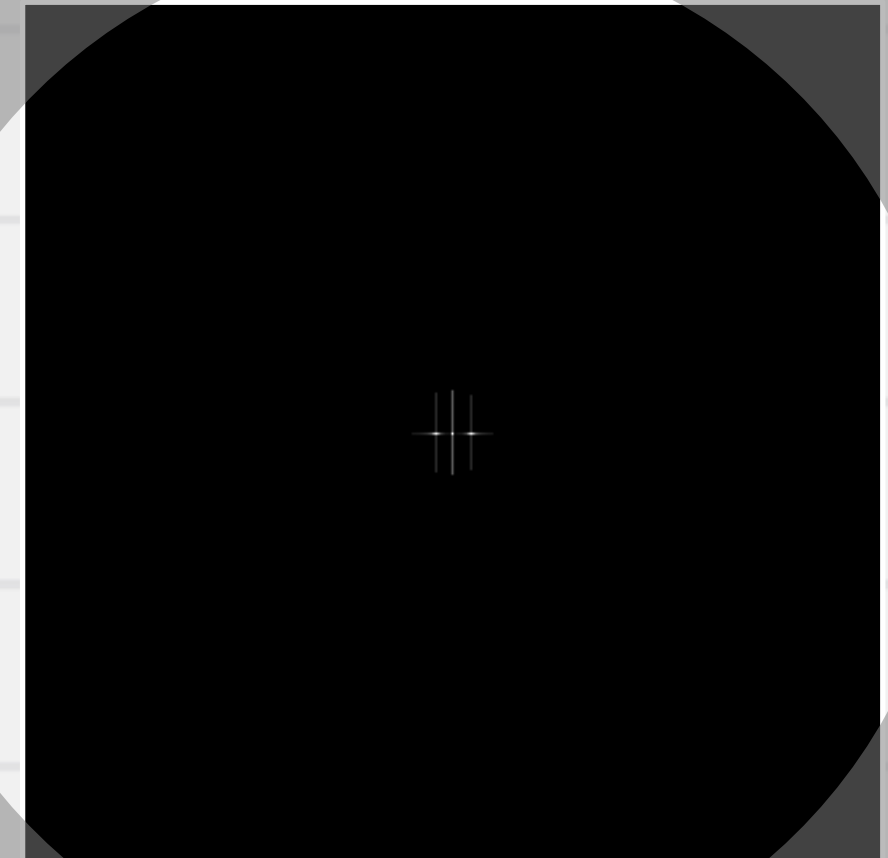
Fast Fourier Transform

Now let's think how can we use that?

Sine Wave Image



Forward Fourier Sine LP filter



That's what will happen

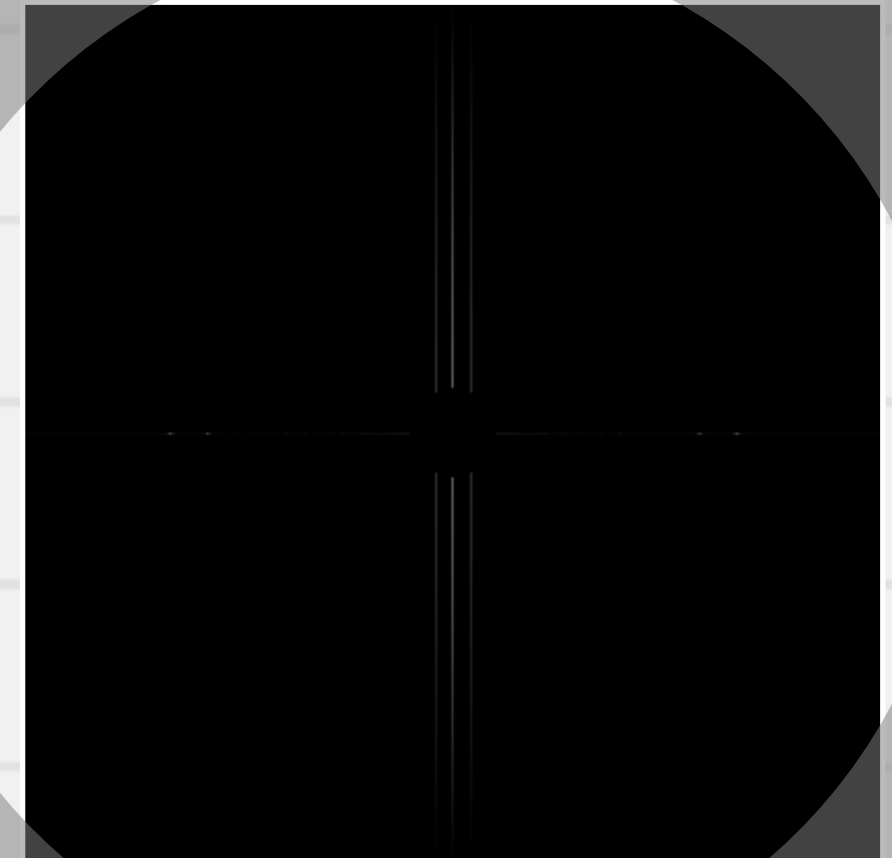
Fast Fourier Transform

Now let's think how can we use that?

Sine Wave Image



Forward Fourier Sine HP filter



And there is something can be more interesting. That if applied high pass filter and instead of keeping image center . We will remove the center and keep all the surrounding pixels.

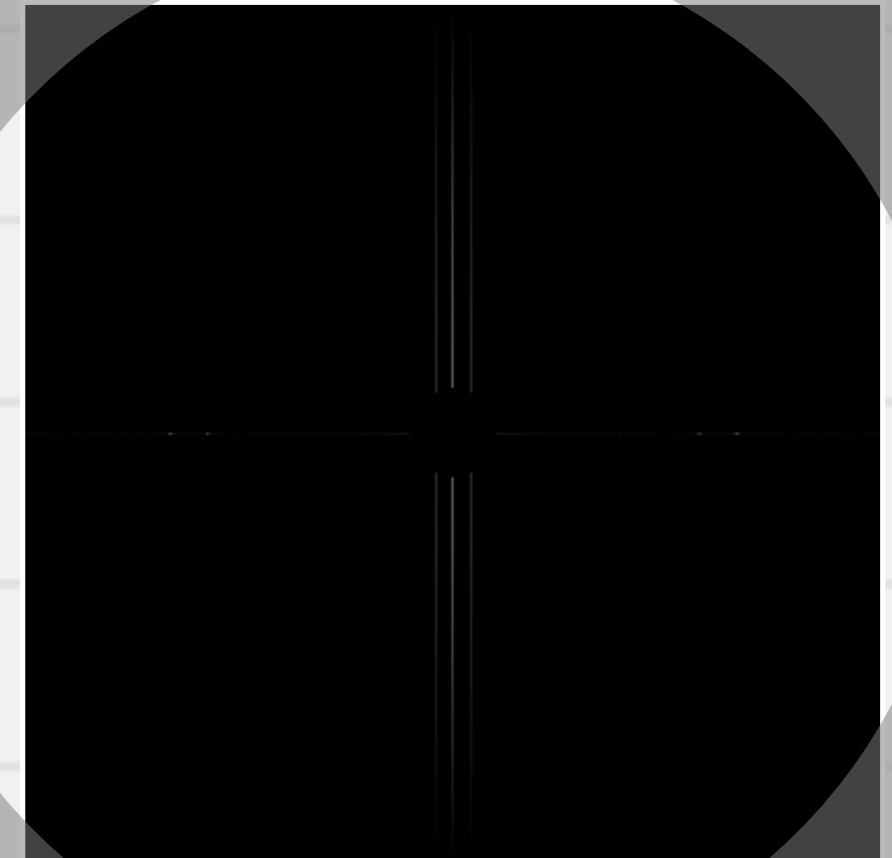
Fast Fourier Transform

Now let's think how can we use that?

Sine Wave Image



Forward Fourier Sine HP filter



And this will sharpen the image edges
instead of blurring it.

Fast Fourier Transform

Source & References

All the previous content was almost copied from:

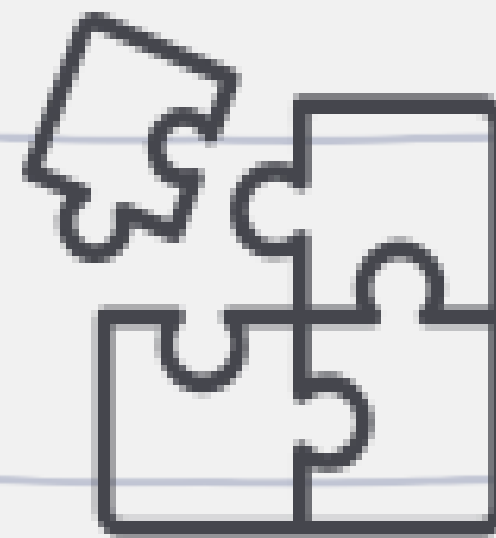
- ◆ Youtube -First Principles of Computer Vision

Author: Prof. Shree Nayar

Published on: Mar 3, 2021



04

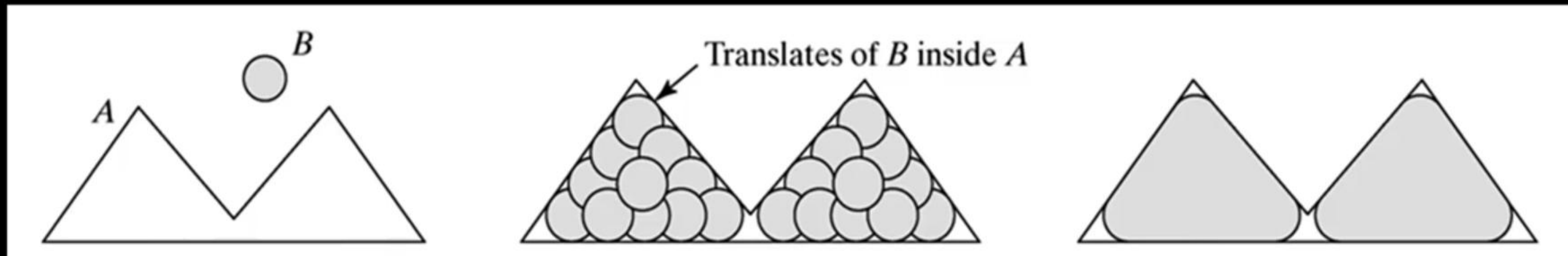


Opening & Closing

Opening

Opening is a morphological operation used to remove small objects or noise from a binary image while preserving the overall shape and size of larger objects. It is done in two steps:

1. Erosion – shrinks the white regions.
2. Dilation – grows the remaining white regions back to their original size.



Binary image A
and structuring
element B

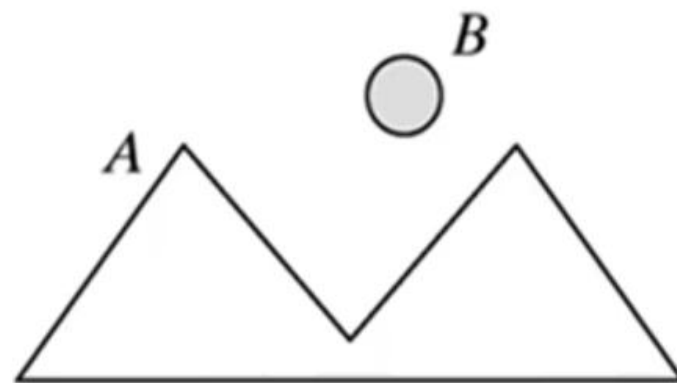
Translations of B
that fit entirely
within A

The opening of A
by B is shown
shaded

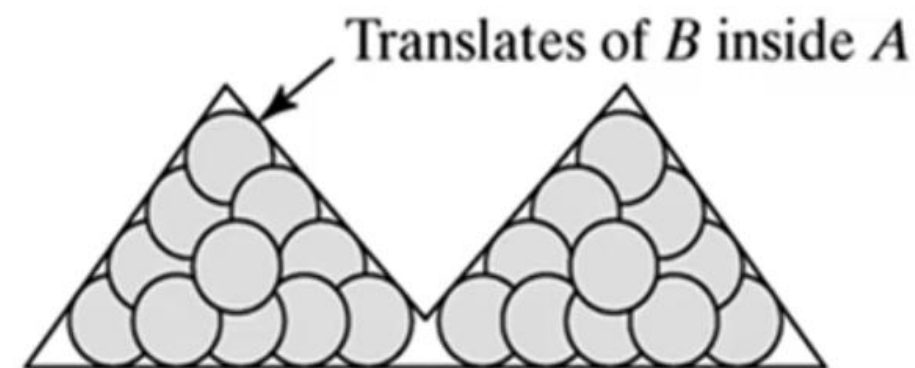
Opening

🧠 Use case:

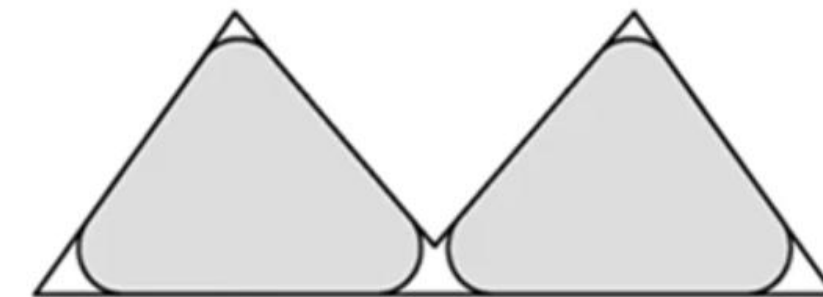
- Removes small noise dots
- Separates objects that are close together
- Smooths object boundaries



Binary image A
and structuring
element B



Translations of B
that fit entirely
within A



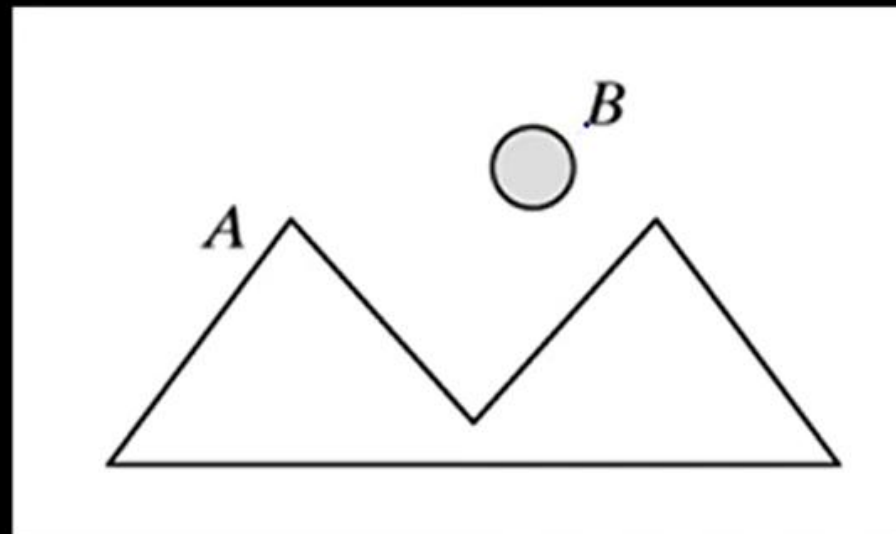
The opening of A
by B is shown
shaded

Closing

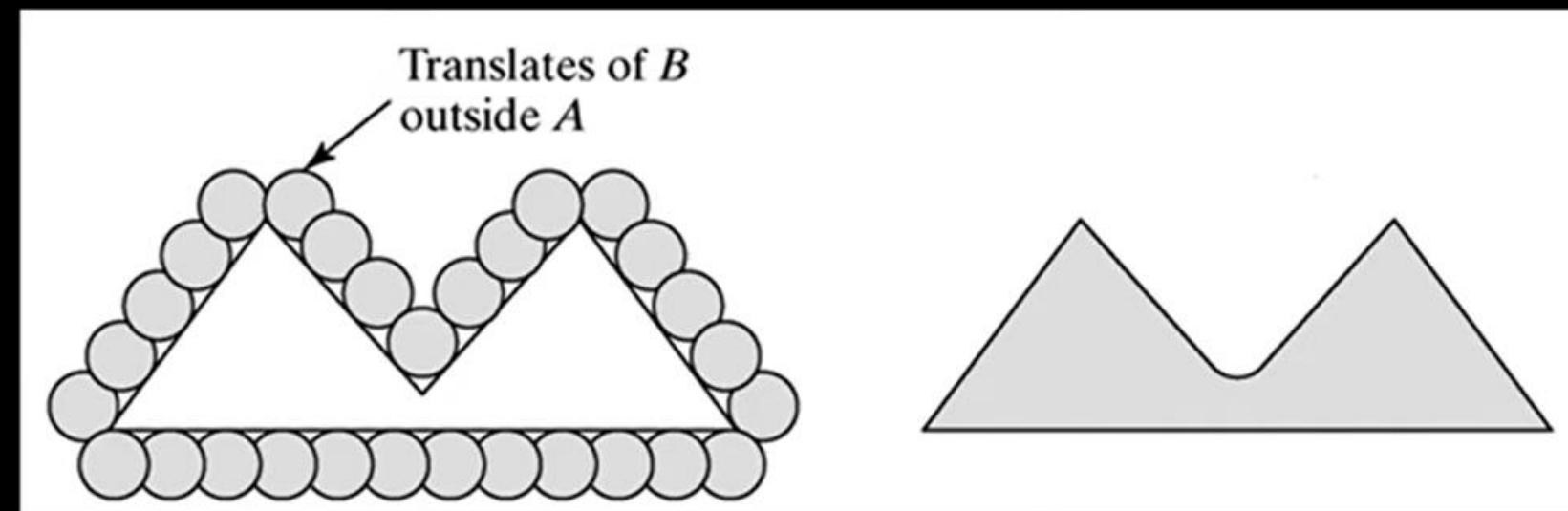
Closing is a morphological operation used to fill small holes or gaps in a binary image while preserving the overall shape and size of the objects.

It is done in two steps:

1. Dilation – expands the white regions.
2. Erosion – shrinks the expanded regions back to original size.



Binary image A and structuring element B



Translations of B that do not overlap A

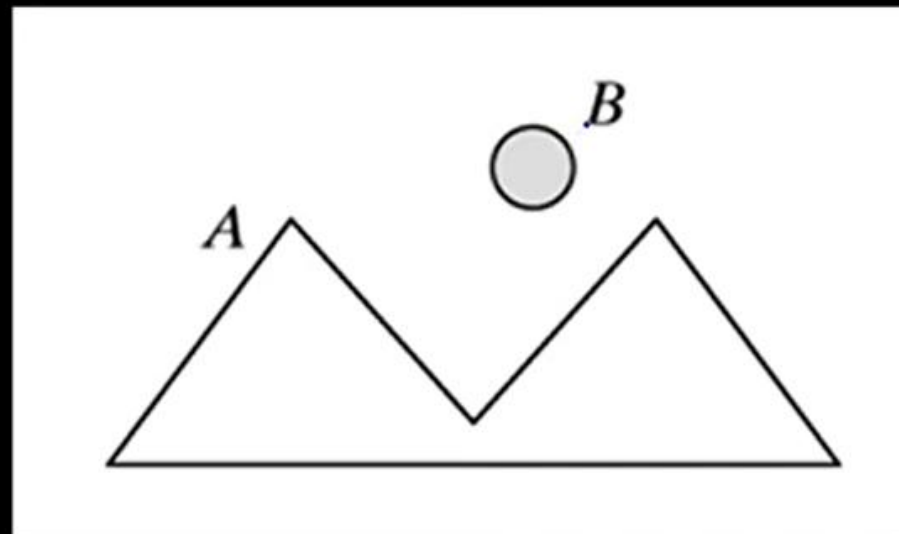


The closing of A by B is shown shaded

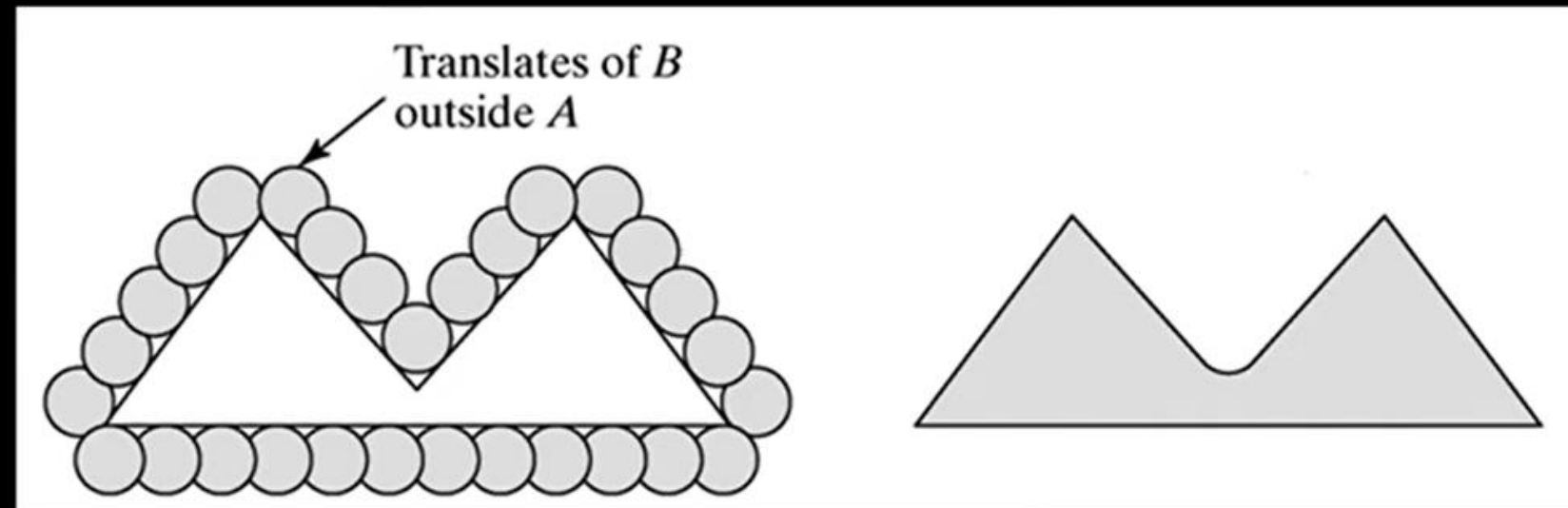
Closing

🧠 Use case:

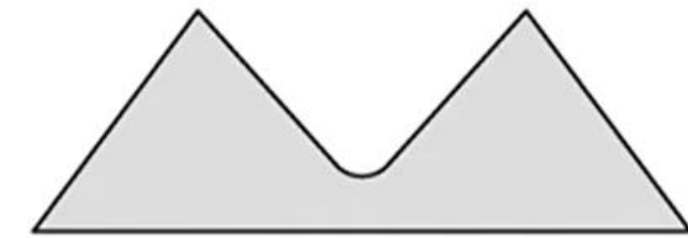
- Fills small black holes inside objects
- Connects small breaks or gaps
- Smooths object contours



Binary image A and structuring element B



Translations of B that do not overlap A



The closing of A by B is shown shaded

Opening & Closing



Opening Followed by Closing

When you apply Opening first, then Closing, you're combining the strengths of both operations to clean the image from both noise and small gaps.



Original image



Opening



Closing



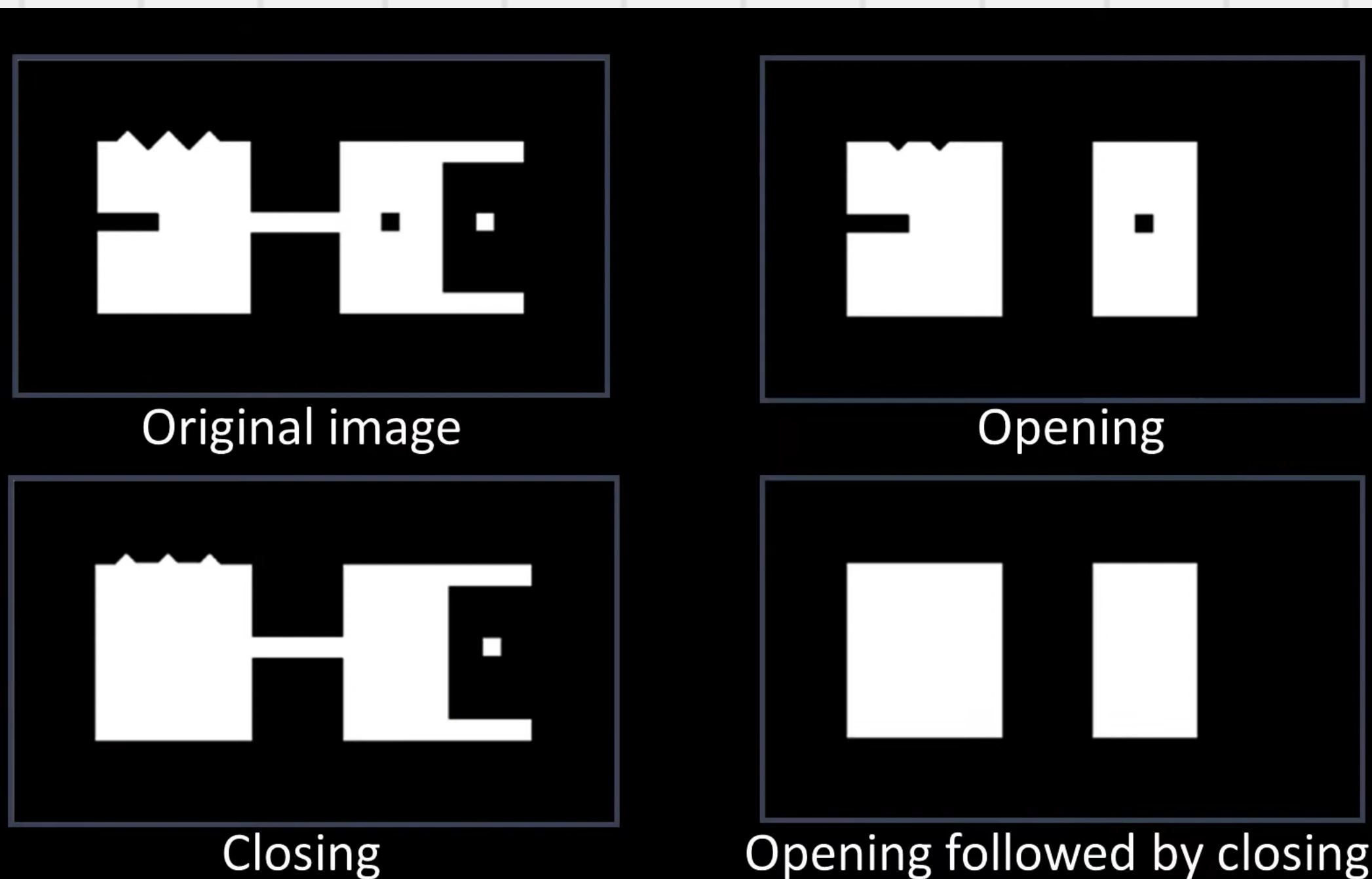
Opening followed by closing

Opening & Closing

- ◆ Step-by-step effect:

1. Opening:

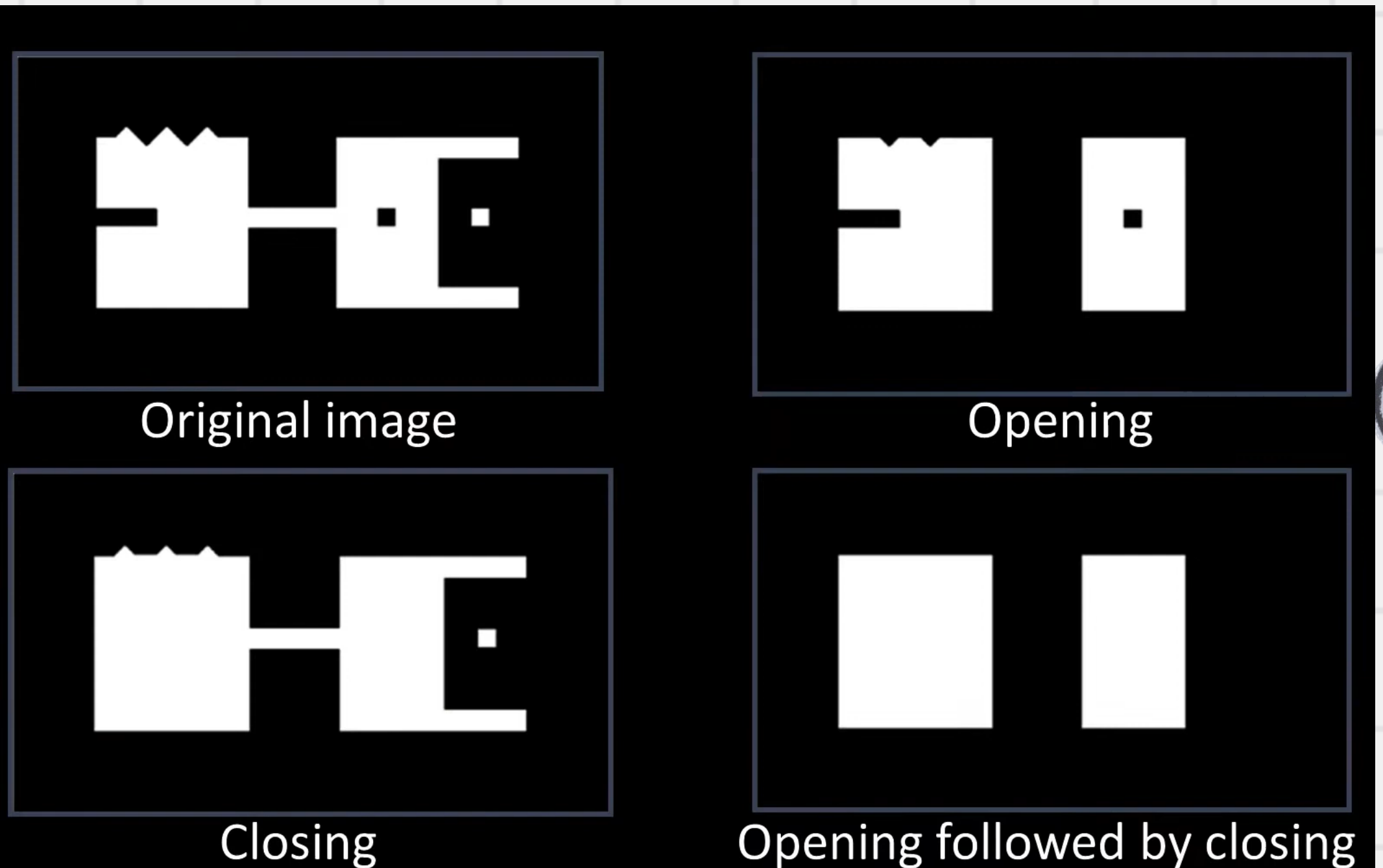
- Removes small white noise (tiny bright dots).
- Detaches small unwanted bright objects.
- Slightly shrinks the objects.



Opening & Closing

2. Then Closing:

- Fills small black holes (tiny dark spots inside objects).
- Connects small gaps or cracks in objects.
- Restores object size without bringing back noise.



Opening & Closing

Dilation & Erosion vs Opening & Closing

Feature	Dilation & Erosion	Opening & Closing
Basic Function	Directly expands or shrinks objects	Combination of erosion & dilation (or reverse)
Pros	✓ Simple & fast ✓ Good for basic shape size control	✓ Removes small noise ✓ Fills tiny gaps without major shape change
Cons	✗ May distort shapes ✗ Doesn't handle noise well	✗ Slightly more complex ✗ Needs 2 steps (open or close)
Use Case	Resize objects, adjust blobs	Preprocessing for object detection, OCR, etc.
Noise Handling	✗ Not ideal	✓ Very effective
Hole Filling	✗ Doesn't fill holes	✓ Closing fills holes inside shapes

Dilation & Erosion

Source & References

The pics in the previous section are directly taken from:

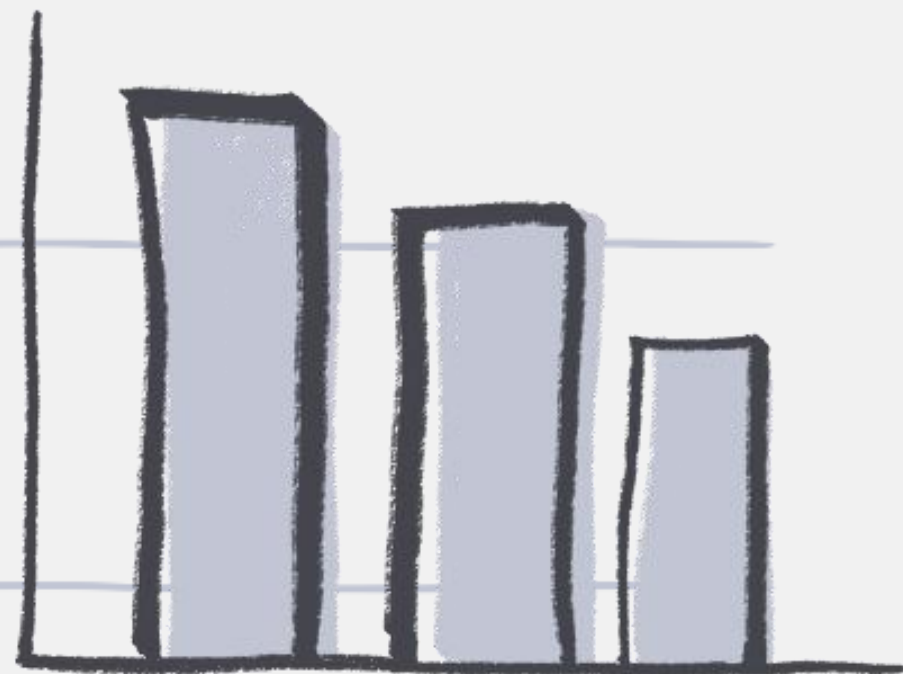
- ◆ Youtube – Udacity “Introduction to Computer Vision Course”

Author: Sebastian Thrun

Published on: Feb 23, 2015

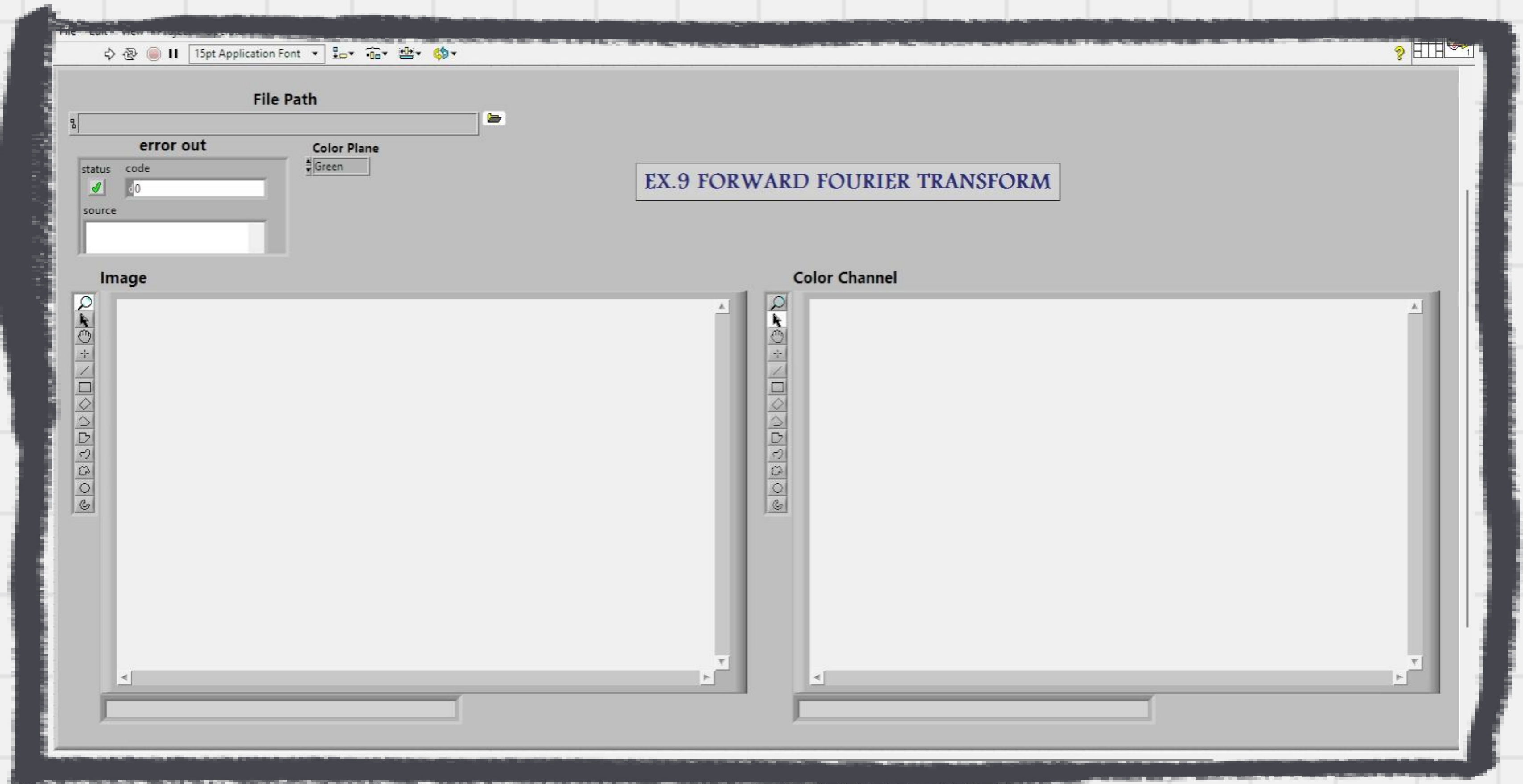


04

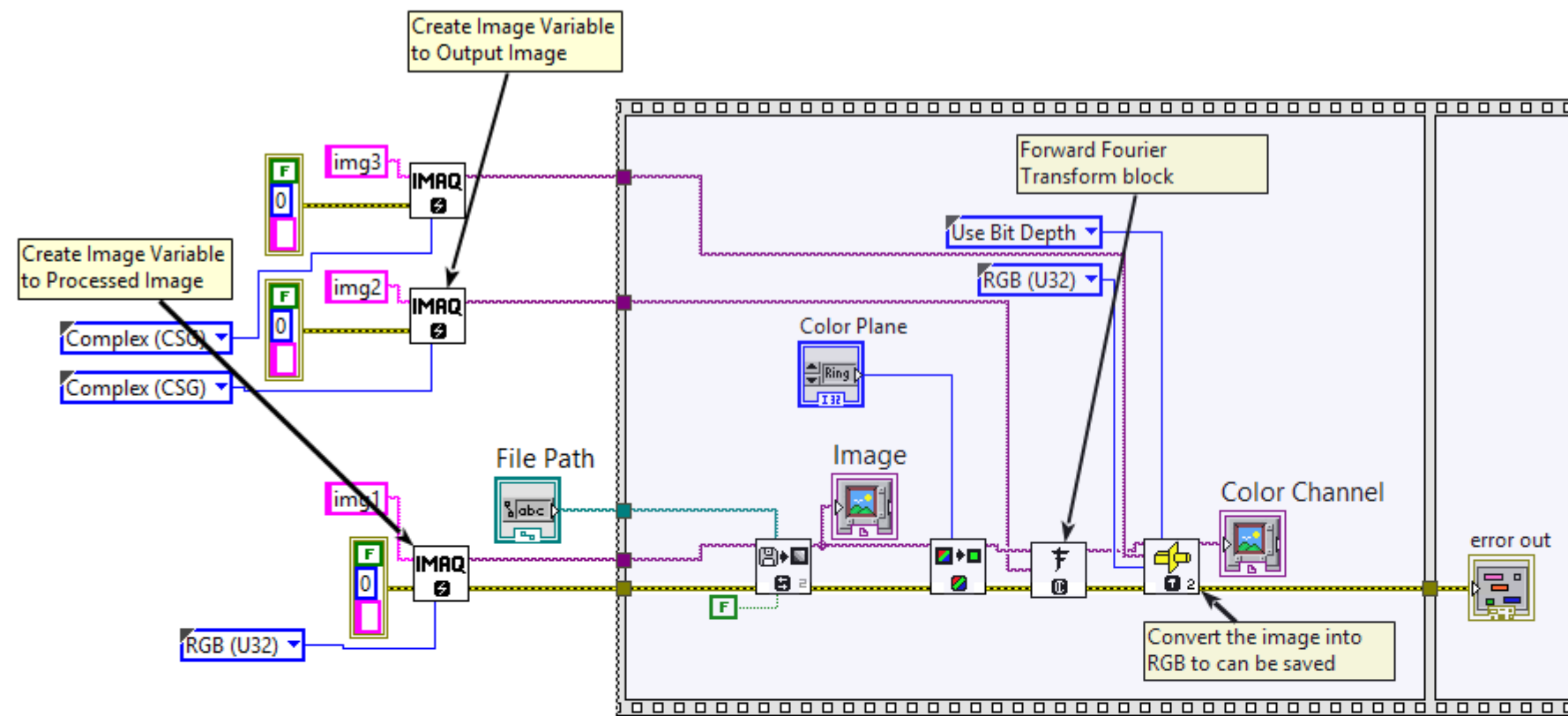


LabVIEW

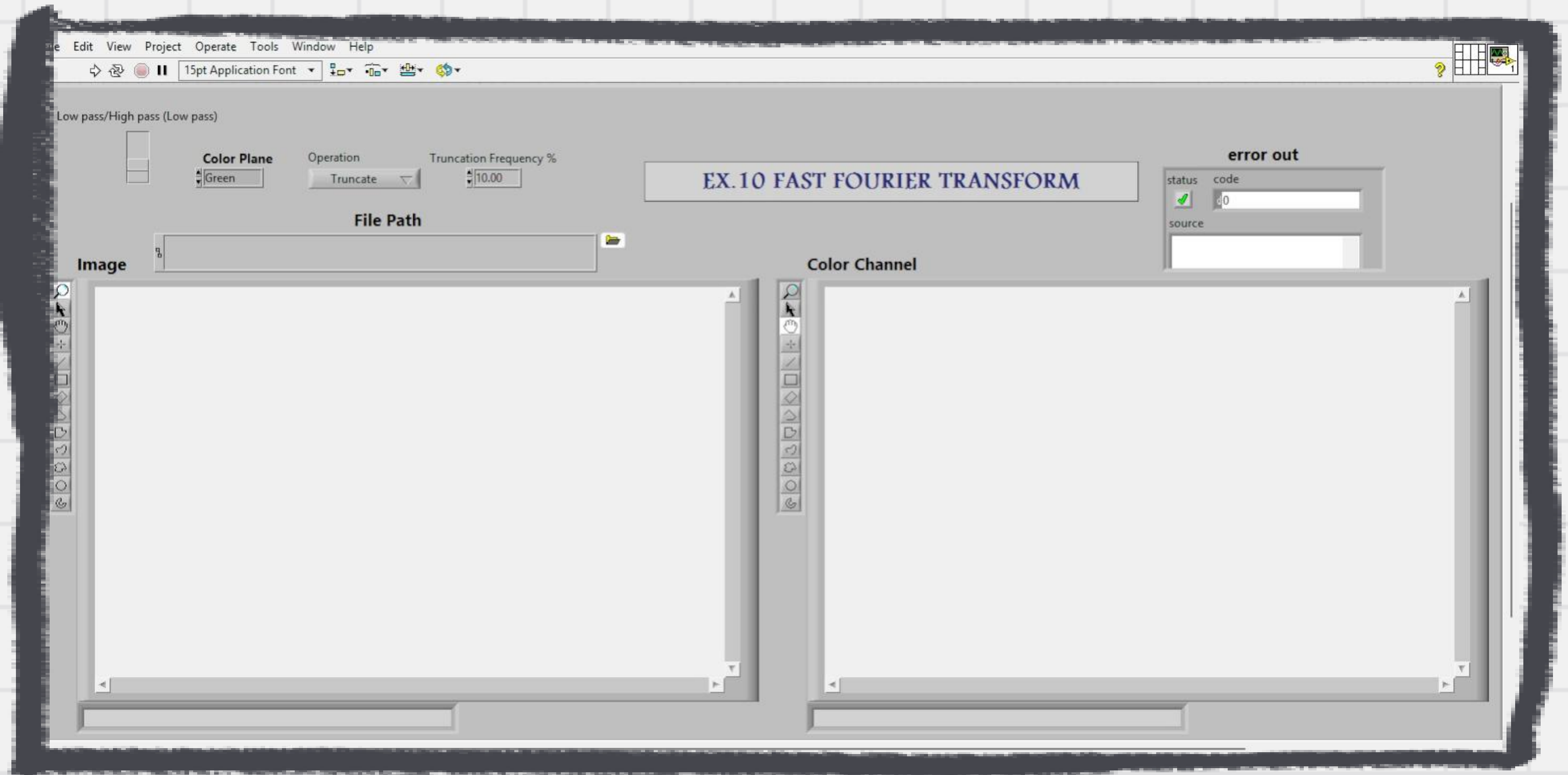
EX9. Forward Fourier Transformation



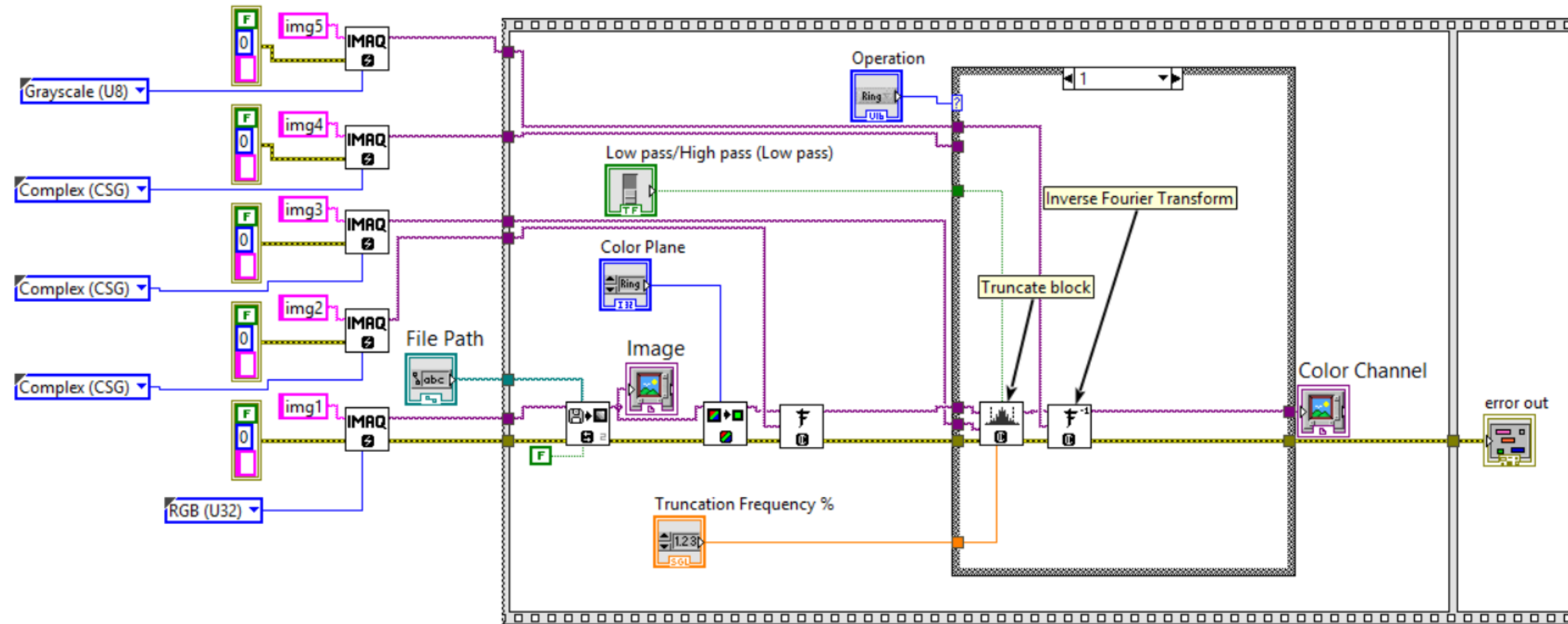
EX9. Forward Fourier Transformation



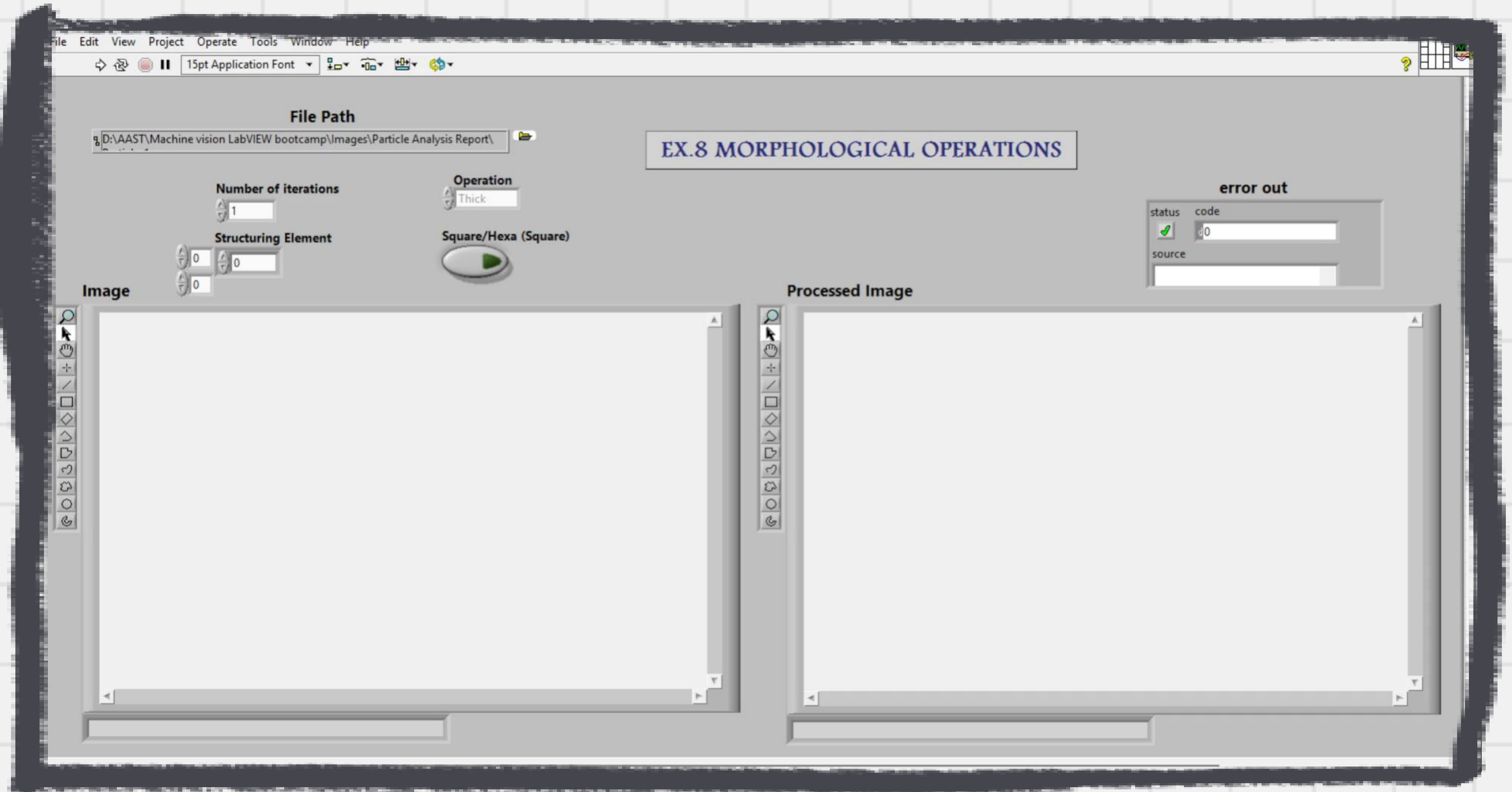
EX10. Forward Fourier Transformation



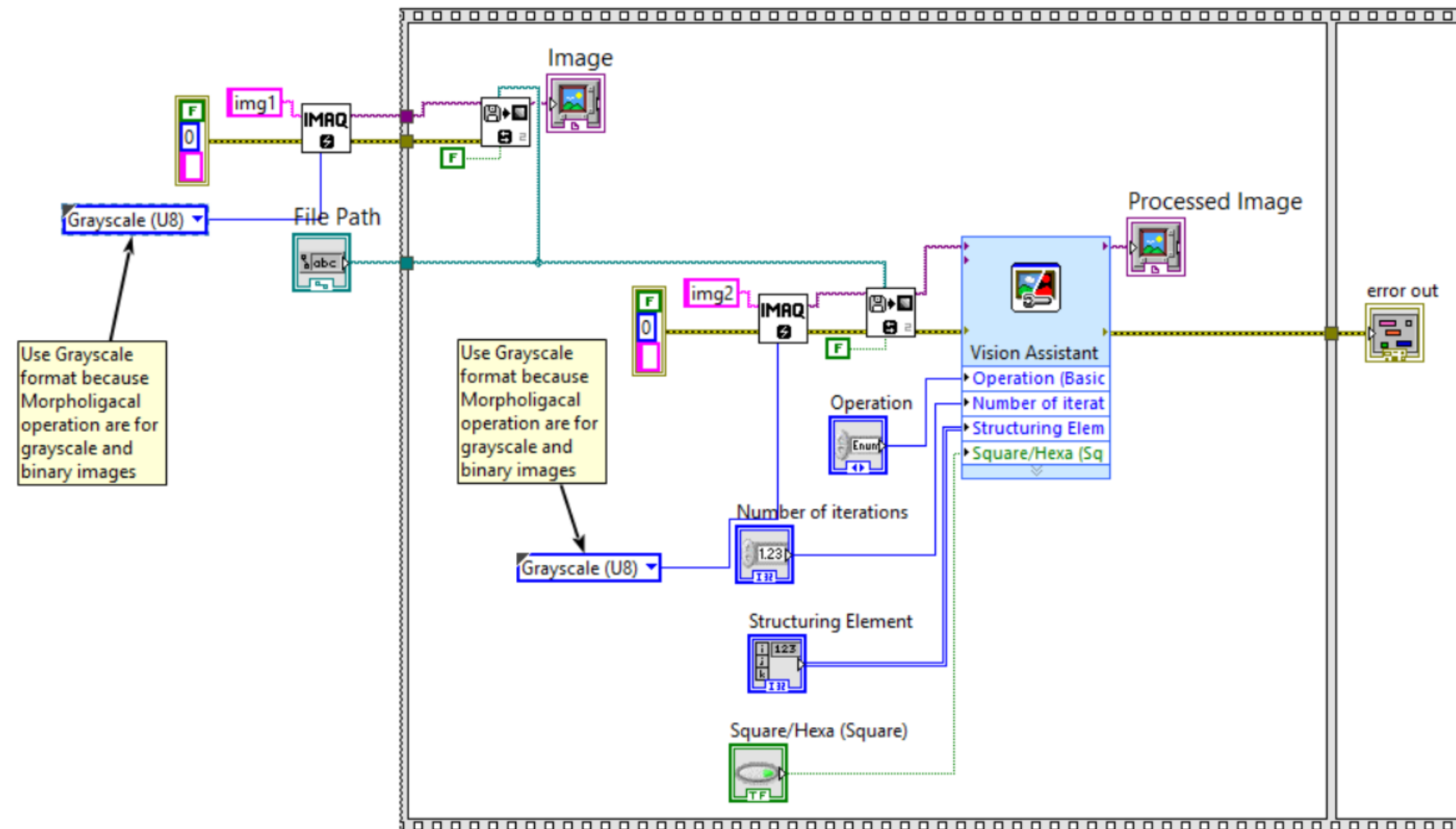
EX10. Forward Fourier Transformation



EX8. Morphological Operations



EX8. Morphological Operations





A hand-drawn sign with a thick black border and a small tab at the bottom right. The sign is set against a light gray grid background. The word "THANK" is written in a bold, dark blue, hand-drawn font and is enclosed in a light blue oval. The word "you" is written below it in the same font, with a light blue horizontal brushstroke underneath. The entire sign has a slightly textured, hand-painted appearance.

THANK

you